

The corresponding minimum MSE is

$$\begin{aligned}\text{MMSE}_2 &= 1 - h(0)\gamma_{ss}(0) - h(1)\gamma_{ss}(1) \\ &= 1 - 0.451 - (0.165)(0.6) \\ &= 0.45\end{aligned}$$

This error can be reduced further by increasing the length of the Wiener filter (see Problem 11.27).

11.6.2 Orthogonality Principle in Linear Mean-Square Estimation

The normal equations for the optimum filter coefficients given by (11.6.3) can be obtained directly by applying the orthogonality principle in linear mean-square estimation. Simply stated, the mean-square error $\mathcal{E}_M$ in (11.6.2) is a minimum if the filter coefficients $\{h(k)\}$ are selected such that the error is orthogonal to each of the data points in the estimate,

$$E[e(n)x^*(n-l)] = 0 \quad l = 0, 1, \dots, M-1 \quad (11.6.12)$$

where

$$e(n) = d(n) - \sum_{k=0}^{M-1} h(k)x(n-k) \quad (11.6.13)$$

Conversely, if the filter coefficients satisfy (11.6.12), the resulting MSE is a minimum.

When viewed geometrically, the output of the filter, which is the estimate

$$\hat{d}(n) = \sum_{k=0}^{M-1} h(k)x(n-k) \quad (11.6.14)$$

is a vector in the subspace spanned by the data $\{x(k), 0 \leq k \leq M-1\}$. The error $e(n)$ is a vector from $d(n)$ to $\hat{d}(n)$ [i.e., $d(n) = e(n) + \hat{d}(n)$], as shown in Fig. 11.9. The orthogonality principle states that the length $\mathcal{E}_M = E|e(n)|^2$ is a minimum when $e(n)$ is perpendicular to the data subspace [i.e., $e(n)$ is orthogonal to each data point $x(k), 0 \leq k \leq M-1$].

We note that the solution obtained from the normal equations in (11.6.3) is unique if the data $\{x(n)\}$ in the estimate $\hat{d}(n)$ are *linearly independent*. In this case, the correlation matrix Γ_M is nonsingular. On the other hand, if the data are linearly dependent, the rank of Γ_M is less than M and therefore the solution is not unique. In this case, the estimate $\hat{d}(n)$ can be expressed as a linear combination of a reduced set of linearly independent data points equal to the rank of Γ_M .

Since the MSE is minimized by selecting the filter coefficients to satisfy the orthogonality principle, the residual minimum MSE is simply

$$\text{MMSE}_M = E[e(n)d^*(n)] \quad (11.6.15)$$

which yields the result given in (11.6.6).

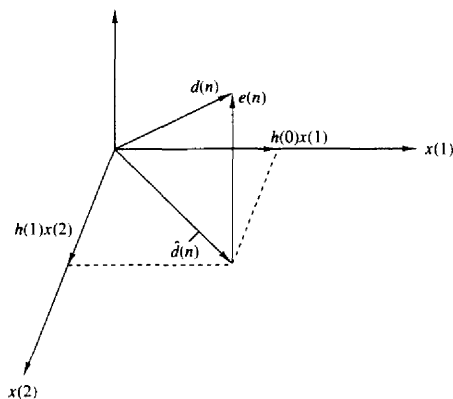


Figure 11.9 Geometric interpretation of linear MSE problem

11.6.3 IIR Wiener Filter

In the preceding section we constrained the filter to be FIR and obtained a set of M linear equations for the optimum filter coefficients. In this section we allow the filter to be infinite in duration (IIR) and the data sequence to be infinite as well. Hence the filter output is

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) \quad (11.6.16)$$

The filter coefficients are selected to minimize the mean-square error between the desired output $d(n)$ and $y(n)$, that is,

$$\begin{aligned} \mathcal{E}_{\infty} &= E|e(n)|^2 \\ &= E \left| d(n) - \sum_{k=0}^{\infty} h(k)x(n-k) \right|^2 \end{aligned} \quad (11.6.17)$$

Application of the orthogonality principle leads to the Wiener-Hopf equation,

$$\sum_{k=0}^{\infty} h(k)\gamma_{xx}(l-k) = \gamma_{dx}(l) \quad l \geq 0 \quad (11.6.18)$$

The residual MMSE is simply obtained by application of the condition given by (11.6.15). Thus we obtain

$$\text{MMSE}_{\infty} = \min_{\mathbf{h}} \mathcal{E}_{\infty} = \sigma_d^2 - \sum_{k=0}^{\infty} h_{\text{opt}}(k)\gamma_{dx}^*(k) \quad (11.6.19)$$

The Wiener-Hopf equation given by (11.6.18) cannot be solved directly with z -transform techniques, because the equation holds only for $l \geq 0$. We shall solve

for the optimum IIR Wiener filter based on the innovations representation of the stationary random process $\{x(n)\}$.

Recall that a stationary random process $\{x(n)\}$ with autocorrelation $\gamma_{xx}(k)$ and power spectral density $\Gamma_{xx}(f)$ can be represented by an equivalent innovations process, $\{i(n)\}$ by passing $\{x(n)\}$ through a noise-whitening filter with system function $1/G(z)$, where $G(z)$ is the minimum-phase part obtained from the spectral factorization of $\Gamma_{xx}(z)$:

$$\Gamma_{xx}(z) = \sigma_i^2 G(z)G(z^{-1}) \quad (11.6.20)$$

Hence $G(z)$ is analytic in the region $|z| > r_1$, where $r_1 < 1$.

Now, the optimum Wiener filter can be viewed as the cascade of the whitening filter $1/G(z)$ with a second filter, say $Q(z)$, whose output $y(n)$ is identical to the output of the optimum Wiener filter. Since

$$y(n) = \sum_{k=0}^{\infty} q(k)i(n-k) \quad (11.6.21)$$

and $e(n) = d(n) - y(n)$, application of the orthogonality principle yields the new Wiener-Hopf equation as

$$\sum_{k=0}^{\infty} q(k)\gamma_{di}(l-k) = \gamma_{di}(l) \quad l \geq 0 \quad (11.6.22)$$

But since $\{i(n)\}$ is white, it follows that $\gamma_{di}(l-k) = 0$ unless $l = k$. Thus we obtain the solution as

$$q(l) = \frac{\gamma_{di}(l)}{\gamma_{ii}(0)} = \frac{\gamma_{di}(l)}{\sigma_i^2} \quad l \geq 0 \quad (11.6.23)$$

The z -transform of the sequence $\{q(l)\}$ is

$$\begin{aligned} Q(z) &= \sum_{k=0}^{\infty} q(k)z^{-k} \\ &= \frac{1}{\sigma_i^2} \sum_{k=0}^{\infty} \gamma_{di}(k)z^{-k} \end{aligned} \quad (11.6.24)$$

If we denote the z -transform of the two-sided crosscorrelation sequence $\gamma_{di}(k)$ by $\Gamma_{di}(z)$:

$$\Gamma_{di}(z) = \sum_{k=-\infty}^{\infty} \gamma_{di}(k)z^{-k} \quad (11.6.25)$$

and define $[\Gamma_{di}(z)]_+$ as

$$[\Gamma_{di}(z)]_+ = \sum_{k=0}^{\infty} \gamma_{di}(k)z^{-k} \quad (11.6.26)$$

then

$$Q(z) = \frac{1}{\sigma_i^2} [\Gamma_{di}(z)]_+ \quad (11.6.27)$$

To determine $[\Gamma_{di}(z)]_+$, we begin with the output of the noise-whitening filter, which can be expressed as

$$i(n) = \sum_{k=0}^{\infty} v(k)x(n-k) \quad (11.6.28)$$

where $\{v(k), k \geq 0\}$ is the impulse response of the noise-whitening filter,

$$\frac{1}{G(z)} \equiv V(z) = \sum_{k=0}^{\infty} v(k)z^{-k} \quad (11.6.29)$$

Then

$$\begin{aligned} \gamma_{di}(k) &= E[d(n)i^*(n-k)] \\ &= \sum_{m=0}^{\infty} v(m)E[d(n)x^*(n-m-k)] \\ &= \sum_{m=0}^{\infty} v(m)\gamma_{dx}(k+m) \end{aligned} \quad (11.6.30)$$

The z -transform of the crosscorrelation $\gamma_{di}(k)$ is

$$\begin{aligned} \Gamma_{di}(z) &= \sum_{k=-\infty}^{\infty} \left[\sum_{m=0}^{\infty} v(m)\gamma_{dx}(k+m) \right] z^{-k} \\ &= \sum_{m=0}^{\infty} v(m) \sum_{k=-\infty}^{\infty} \gamma_{dx}(k+m)z^{-k} \\ &= \sum_{m=0}^{\infty} v(m)z^m \sum_{k=-\infty}^{\infty} \gamma_{dx}(k)z^{-k} \\ &= V(z^{-1})\Gamma_{dx}(z) = \frac{\Gamma_{dx}(z)}{G(z^{-1})} \end{aligned} \quad (11.6.31)$$

Therefore,

$$Q(z) = \frac{1}{\sigma_i^2} \left[\frac{\Gamma_{dx}(z)}{G(z^{-1})} \right]_+ \quad (11.6.32)$$

Finally, the optimum IIR Wiener filter has the system function

$$\begin{aligned} H_{\text{opt}}(z) &= \frac{Q(z)}{G(z)} \\ &= \frac{1}{\sigma_i^2 G(z)} \left[\frac{\Gamma_{dx}(z)}{G(z^{-1})} \right]_+ \end{aligned} \quad (11.6.33)$$

In summary, the solution for the optimum IIR Wiener filter requires that we perform a spectral factorization of $\Gamma_{xx}(z)$ to obtain $G(z)$, the minimum-phase component, and then we solve for the causal part of $\Gamma_{dx}(z)/G(z^{-1})$. The following example illustrates the procedure.

Example 11.6.2

Let us determine the optimum IIR Wiener filter for the signal given in Example 11.6.1.

Solution For this signal we have

$$\Gamma_{xx}(z) = \Gamma_{ss}(z) + 1 = \frac{1.8(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{3}z)}{(1 - 0.6z^{-1})(1 - 0.6z)}$$

where $\sigma_v^2 = 1.8$ and

$$G(z) = \frac{1 - \frac{1}{3}z^{-1}}{1 - 0.6z^{-1}}$$

The z -transform of the crosscorrelation $\gamma_{dx}(m)$ is

$$\Gamma_{dx}(z) = \Gamma_{sx}(z) = \frac{0.64}{(1 - 0.6z^{-1})(1 - 0.6z)}$$

Hence

$$\begin{aligned} \left[\frac{\Gamma_{dx}(z)}{G(z^{-1})} \right]_+ &= \left[\frac{0.64}{(1 - \frac{1}{3}z)(1 - 0.6z^{-1})} \right]_+ \\ &= \left[\frac{0.8}{1 - 0.6z^{-1}} + \frac{0.266z}{1 - \frac{1}{3}z} \right]_+ \\ &= \frac{0.8}{1 - 0.6z^{-1}} \end{aligned}$$

The optimum IIR filter has the system function

$$\begin{aligned} H_{\text{opt}}(z) &= \frac{1}{1.8} \left(\frac{1 - 0.6z^{-1}}{1 - \frac{1}{3}z^{-1}} \right) \left(\frac{0.8}{1 - 0.6z^{-1}} \right) \\ &= \frac{\frac{4}{9}}{1 - \frac{1}{3}z^{-1}} \end{aligned}$$

and an impulse response

$$h_{\text{opt}}(n) = \frac{4}{9} \left(\frac{1}{3} \right)^n \quad n \geq 0$$

We conclude this section by expressing the minimum MSE given by (11.6.19) in terms of the frequency-domain characteristics of the filter. First, we note that $\sigma_d^2 \equiv E|d(n)|^2$ is simply the value of the autocorrelation sequence $\{\gamma_{dd}(k)\}$ evaluated at $k = 0$. Since

$$\gamma_{dd}(k) = \frac{1}{2\pi j} \oint_{\mathcal{C}} \Gamma_{dd}(z) z^{k-1} dz \quad (11.6.34)$$

it follows that

$$\sigma_d^2 = \gamma_{dd}(0) = \frac{1}{2\pi j} \oint_{\mathcal{C}} \frac{\Gamma_{dd}(z)}{z} dz \quad (11.6.35)$$

where the contour integral is evaluated along a closed path encircling the origin in the region of convergence of $\Gamma_{dd}(z)$.

The second term in (11.6.19) is also easily transformed to the frequency domain by application of Parseval's theorem. Since $h_{\text{opt}}(k) = 0$ for $k < 0$, we have

$$\sum_{k=-\infty}^{\infty} h_{\text{opt}}(k) \gamma_{dx}^*(k) = \frac{1}{2\pi j} \oint_C H_{\text{opt}}(z) \Gamma_{dx}(z^{-1}) z^{-1} dz \quad (11.6.36)$$

where C is a closed contour encircling the origin that lies within the common region of convergence of $H_{\text{opt}}(z)$ and $\Gamma_{dx}(z^{-1})$.

By combining (11.6.35) with (11.6.36), we obtain the desired expression for the MMSE_{∞} in the form

$$\text{MMSE}_{\infty} = \frac{1}{2\pi j} \oint_C [\Gamma_{dd}(z) - H_{\text{opt}}(z) \Gamma_{dx}(z^{-1})] z^{-1} dz \quad (11.6.37)$$

Example 11.6.3

For the optimum Wiener filter derived in Example 11.6.2, the minimum MSE is

$$\text{MMSE}_{\infty} = \frac{1}{2\pi j} \oint_C \left[\frac{0.3555}{(z - \frac{1}{3})(1 - 0.6z)} \right] dz$$

There is a single pole inside the unit circle at $z = \frac{1}{3}$. By evaluating the residue at the pole, we obtain

$$\text{MMSE}_{\infty} = 0.444$$

We observe that this MMSE is only slightly smaller than that for the optimum two-tap Wiener filter in Example 11.6.1.

11.6.4 Noncausal Wiener Filter

In the preceding section we constrained the optimum Wiener filter to be causal [i.e., $h_{\text{opt}}(n) = 0$ for $n < 0$]. In this section we drop this condition and allow the filter to include both the infinite past and the infinite future of the sequence $\{x(n)\}$ in forming the output $y(n)$, that is,

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \quad (11.6.38)$$

The resulting filter is physically unrealizable. It can also be viewed as a *smoothing filter* in which the infinite future signal values are used to smooth the estimate $\hat{d}(n) = y(n)$ of the desired signal $d(n)$.

Application of the orthogonality principle yields the Wiener-Hopf equation for the noncausal filter in the form

$$\sum_{k=-\infty}^{\infty} h(k) \gamma_{xx}(l-k) = \gamma_{dx}(l) \quad -\infty < l < \infty \quad (11.6.39)$$

and the resulting MMSE_{nc} as

$$\text{MMSE}_{nc} = \sigma_d^2 - \sum_{k=-\infty}^{\infty} h(k) \gamma_{dx}^*(k) \quad (11.6.40)$$

Since (11.6.39) holds for $-\infty < l < \infty$, this equation can be transformed directly to yield the optimum noncausal Wiener filter as

$$H_{nc}(z) = \frac{\Gamma_{dx}(z)}{\Gamma_{xx}(z)} \quad (11.6.41)$$

The MMSE_{nc} can also be simply expressed in the z -domain as

$$\text{MMSE}_{nc} = \frac{1}{2\pi j} \oint_C [\Gamma_{dd}(z) - H_{nc}(z)\Gamma_{dx}(z^{-1})] z^{-1} dz \quad (11.6.42)$$

In the following example we compare the form of the optimal noncausal filter with the optimal causal filter obtained in the previous section.

Example 11.6.4

The optimum noncausal Wiener filter for the signal characteristics given in Example 11.6.1 is given by (11.6.41), where

$$\Gamma_{dx}(z) = \Gamma_{sx}(z) = \frac{0.64}{(1 - 0.6z^{-1})(1 - 0.6z)}$$

and

$$\begin{aligned} \Gamma_{xx}(z) &= \Gamma_{ss}(z) + 1 \\ &= \frac{2(1 - 0.3z^{-1} - 0.3z)}{(1 - 0.6z^{-1})(1 - 0.6z)} \end{aligned}$$

Then,

$$H_{nc}(z) = \frac{0.3555}{(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{3}z)}$$

This filter is clearly noncausal.

The minimum MSE achieved by this filter is determined from evaluating (11.6.42). The integrand is

$$\frac{1}{z} \Gamma_{ss}(z) [1 - H_{nc}(z)] = \frac{0.3555}{(z - \frac{1}{3})(1 - \frac{1}{3}z)}$$

The only pole inside the unit circle is $z = \frac{1}{3}$. Hence the residue is

$$\left. \frac{0.3555}{1 - \frac{1}{3}z} \right|_{z=\frac{1}{3}} = \frac{0.3555}{8/9} = 0.40$$

Hence the minimum achievable MSE obtained with the optimum noncausal Wiener filter is

$$\text{MMSE}_{nc} = 0.40$$

Note that this is lower than the MMSE for the causal filter, as expected.

11.7 SUMMARY AND REFERENCES

The major focal point in this chapter is the design of optimum linear systems for linear prediction and filtering. The criterion for optimality is the minimization of the mean-square error between a specified desired filter output and the actual filter output.

In the development of linear prediction, we demonstrated that the equations for the forward and backward prediction errors specified a lattice filter whose parameters, the reflection coefficients $\{K_m\}$, were simply related to the filter coefficients $\{a_m(k)\}$ of the direct form FIR linear predictor and the associated prediction error filter. The optimum filter coefficients $\{K_m\}$ and $\{a_m(k)\}$ are easily obtained from the solution of the normal equations.

We described two computationally efficient algorithms for solving the normal equations, the Levinson–Durbin algorithm and the Schür algorithm. Both algorithms are suitable for solving a Toeplitz system of linear equations and have a computational complexity of $O(p^2)$ when executed on a single processor. However, with full parallel processing, the Schür algorithm solves the normal equations in $O(p)$ time, whereas the Levinson–Durbin algorithm requires $O(p \log p)$ time.

In addition to the all-zero lattice filter resulting from linear prediction, we also derived the AR lattice (all-pole) filter structure and the ARMA lattice-ladder (pole-zero) filter structure. Finally, we described the design of the class of optimum linear filters, called Wiener filters.

Linear estimation theory has had a long and rich history of development over the past four decades. Kailath (1974) presents a historical account of the first three decades. The pioneering work of Wiener (1949) on optimum linear filtering for statistically stationary signals is especially significant. The generalization of the Wiener filter theory to dynamical systems with random inputs was developed by Kalman (1960) and Kalman and Bucy (1961). Kalman filters are treated in the books by Meditch (1969), Brown (1983), and Chui and Chen (1987). The monograph by Kailath (1981) treats both Wiener and Kalman filters.

There are numerous references on linear prediction and lattice filters. Tutorial treatments on these subjects have been published in the journal papers by Makhoul (1975, 1978) and Friedlander (1982a, b). The books by Haykin (1991), Markel and Gray (1976), and Tretter (1976) provide comprehensive treatments of these subjects. Applications of linear prediction to spectral analysis are found in the books by Kay (1988) and Marple (1987), to geophysics in the book Robinson and Treitel (1980), and to adaptive filtering in the book by Haykin (1991).

The Levinson–Durbin algorithm for solving the normal equations recursively was given by Levinson (1947) and later modified by Durbin (1959). Variations of this classical algorithm, called *split Levinson algorithms*, have been developed by Delsarte and Genin (1986) and by Krishna (1988). These algorithms exploit additional symmetries in the Toeplitz correlation matrix and save about a factor of 2 in the number of multiplications.

The Schür algorithm was originally described by Schür (1917) in a paper published in German. An English translation of this paper appears in the book edited by Gohberg (1986). The Schür algorithm is intimately related to the polynomials $\{A_m(z)\}$, which can be interpreted as orthogonal polynomials. A treatment of orthogonal polynomials is given in the books by Szegő (1967), Grenander and Szegő (1958), and Geronimus (1958). The thesis of Vieira (1977) and the papers by Kailath et al. (1978), Delsarte et al. (1978), and Youla and Kazanjian (1978)

provide additional results on orthogonal polynomials. Kailath (1985, 1986) provides tutorial treatments of the Schür algorithm and its relationship to orthogonal polynomials and the Levinson–Durbin algorithm. The pipelined parallel processing structure for computing the reflection coefficients based on the Schür algorithm and the related problem of solving Toeplitz systems of linear equations is described in the paper by Kung and Hu (1983). Finally, we should mention that some additional computational efficiency can be achieved in the Schür algorithm, by further exploiting symmetry properties of Toeplitz matrices, as described by Krishna (1988). This leads to the so-called split-Schür algorithm, which is analogous to the split-Levinson algorithm.

PROBLEMS

11.1 The power density spectrum of an AR process $\{x(n)\}$ is given as

$$\begin{aligned}\Gamma_{xx}(\omega) &= \frac{\sigma_w^2}{|A(\omega)|^2} \\ &= \frac{25}{|1 - e^{-j\omega} + \frac{1}{2}e^{-j2\omega}|^2}\end{aligned}$$

where σ_w^2 is the variance of the input sequence.

- (a) Determine the difference equation for generating the AR process when the excitation is white noise.
- (b) Determine the system function for the whitening filter.

11.2 An ARMA process has an autocorrelation $\{\gamma_{xx}(m)\}$ whose z -transform is given as

$$\Gamma_{xx}(z) = 9 \frac{(z - \frac{1}{3})(z - 3)}{(z - \frac{1}{2})(z - 2)} \quad \frac{1}{2} < |z| < 2$$

- (a) Determine the filter $H(z)$ for generating $\{x(n)\}$ from a white noise input sequence. Is $H(z)$ unique? Explain.
- (b) Determine a stable linear whitening filter for the sequence $\{x(n)\}$.

11.3 Consider the ARMA process generated by the difference equation

$$x(n) = 1.6x(n-1) - 0.63x(n-2) + w(n) + 0.9w(n-1)$$

- (a) Determine the system function of the whitening filter and its poles and zeros.
- (b) Determine the power density spectrum of $\{x(n)\}$.

11.4 Determine the lattice coefficients corresponding to the FIR filter with system function

$$H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

11.5 Determine the reflection coefficients $\{K_m\}$ of the lattice filter corresponding to the FIR filter described by the system function

$$H(z) = A_2(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$$

- 11.6 (a) Determine the zeros and sketch the zero pattern for the FIR lattice filter with reflection coefficients

$$K_1 = \frac{1}{2} \quad K_2 = -\frac{1}{3} \quad K_3 = 1$$

(b) Repeat part (a) but with $K_3 = -1$.

(c) You should have found that the zeros lie on the unit circle. Can this result be generalized? How?

- 11.7 Determine the impulse response of the FIR filter that is described by the lattice coefficients $K_1 = 0.6$, $K_2 = 0.3$, $K_3 = 0.5$, and $K_4 = 0.9$.

- 11.8 In Section 11.2.4 we indicated that the noise-whitening filter $A_p(z)$ for a causal AR(p) process is a forward linear prediction-error filter of order p . Show that the backward linear prediction-error filter of order p is the noise-whitening filter of the corresponding anticausal AR(p) process.

- 11.9 Use the orthogonality principle to determine the normal equations and the resulting minimum MSE for a forward predictor of order p that predicts m samples ($m > 1$) into the future (m -step forward predictor). Sketch the prediction error filter.

- 11.10 Repeat Problem 11.9 for an m -step backward predictor.

- 11.11 Determine a Levinson–Durbin recursive algorithm for solving for the coefficients of a backward prediction-error filter. Use the result to show that coefficients of the forward and backward predictors can be expressed recursively as

$$\begin{aligned} \mathbf{a}_m &= \begin{bmatrix} \mathbf{a}_{m-1} \\ 0 \end{bmatrix} + K_m \begin{bmatrix} \mathbf{b}_{m-1} \\ 1 \end{bmatrix} \\ \mathbf{b}_m &= \begin{bmatrix} \mathbf{b}_{m-1} \\ 0 \end{bmatrix} + K_m^* \begin{bmatrix} \mathbf{a}_{m-1} \\ 1 \end{bmatrix} \end{aligned}$$

- 11.12 The Levinson–Durbin algorithm described in Section 11.3.1 solved the linear equations

$$\Gamma_m \mathbf{a}_m = -\gamma_m$$

where the right-hand side of this equation has elements of the autocorrelation sequence that are also elements of the matrix Γ . Let us consider the more general problem of solving the linear equations

$$\Gamma_m \mathbf{b}_m = \mathbf{c}_m$$

where $\mathbf{c}_m$ is an arbitrary vector. (The vector $\mathbf{b}_m$ is not related to the coefficients of the backward predictor.) Show that the solution to $\Gamma_m \mathbf{b}_m = \mathbf{c}_m$ can be obtained from a *generalized Levinson–Durbin* algorithm which is given recursively as

$$\begin{aligned} b_m(m) &= \frac{c(m) - \gamma_{m-1}^{b*} b_{m-1}}{E_{m-1}^f} \\ b_m(k) &= b_{m-1}(k) - b_m(m) a_{m-1}^*(m-k) \quad \begin{matrix} k = 1, 2, \dots, m-1 \\ m = 1, 2, \dots, p \end{matrix} \end{aligned}$$

where $b_1(1) = c(1)/\gamma_{xx}(0) = c(1)/E_0^f$ and $a_m(k)$ is given by (11.3.17). Thus a second recursion is required to solve the equation $\Gamma_m \mathbf{b}_m = \mathbf{c}_m$.

- 11.13 Use the generalized Levinson–Durbin algorithm to solve the normal equations recursively for the m -step forward and backward predictors.

- 11.14** Show that the transformation

$$\mathbf{V}_m = \begin{bmatrix} 1 & K_m \\ K_m^* & 1 \end{bmatrix}$$

in the Schur algorithm satisfies the special property

$$\mathbf{V}_m \mathbf{J} \mathbf{V}_m^* = (1 - |K_m|^2) \mathbf{J}$$

where

$$\mathbf{J} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Thus $\mathbf{V}_m$ is called a J -rotation matrix. Its role is to rotate or hyperbolate the row of $\mathbf{G}_m$ to lie along the first coordinate direction (Kailath, 1985).

- 11.15** Prove the additional properties (a) through (l) of the prediction-error filters given in Section 11.4.
- 11.16** Extend the additional properties (a) through (l) of the prediction error filters given in Section 11.4 to complex-valued signals.
- 11.17** Determine the reflection coefficient K_3 in terms of the autocorrelations $\{\gamma_{xx}(m)\}$ from the Schür algorithm and compare your result with the expression for K_3 obtained from the Levinson–Durbin algorithm.
- 11.18** Consider an infinite-length ($p = \infty$) one-step forward predictor for a stationary random process $\{x(n)\}$ with a power density spectrum of $\Gamma_{xx}(f)$. Show that the mean-square error of the prediction-error filter can be expressed as

$$E_\infty^f = 2\pi \exp\left\{\int_{-1/2}^{1/2} \ln \Gamma_{xx}(f) df\right\}$$

- 11.19** Determine the output of an infinite-length ($p = \infty$) m -step forward predictor and the resulting mean-square error when the input signal is a first-order autoregressive process of the form

$$x(n) = ax(n-1) + w(n)$$

- 11.20** An AR(3) process $\{x(n)\}$ is characterized by the autocorrelation sequence $\gamma_{xx}(0) = 1$, $\gamma_{xx}(1) = \frac{1}{2}$, $\gamma_{xx}(2) = \frac{1}{8}$, and $\gamma_{xx}(3) = \frac{1}{64}$.
- (a) Use the Schür algorithm to determine the three reflection coefficients K_1 , K_2 , and K_3 .
- (b) Sketch the lattice filter for synthesizing $\{x(n)\}$ from a white noise excitation.

- 11.21** The purpose of this problem is to show that the polynomials $\{A_m(z)\}$, which are the system functions of the forward prediction-error filters of order m , $m = 0, 1, \dots, p$, can be interpreted as orthogonal on the unit circle. Toward this end, suppose that $\Gamma_{xx}(f)$ is the power spectral density of a zero-mean random process $\{x(n)\}$ and let $\{A_m(z)\}$, $m = 0, 1, \dots, p$, be the system functions of the corresponding prediction-error filters. Show that the polynomials $\{A_m(z)\}$ satisfy the orthogonality property

$$\int_{-1/2}^{1/2} \Gamma_{xx}(f) A_m(f) A_n^*(f) df = E_m^f \delta_{mn} \quad m, n = 0, 1, \dots, p$$

- 11.22** Determine the system function of the all-pole filter described by the lattice coefficients $K_1 = 0.6$, $K_2 = 0.3$, $K_3 = 0.5$, and $K_4 = 0.9$.

- 11.23** Determine the parameters and sketch the lattice-ladder filter structure for the system with system function

$$H(z) = \frac{1 - 0.8z^{-1} + 0.15z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

- 11.24** Consider a signal $x(n) = s(n) + w(n)$, where $s(n)$ is an AR(1) process that satisfies the difference equation

$$s(n) = 0.8s(n-1) + v(n)$$

where $\{v(n)\}$ is a white noise sequence with variance $\sigma_v^2 = 0.49$ and $\{w(n)\}$ is a white noise sequence with variance $\sigma_w^2 = 1$. The processes $\{v(n)\}$ and $\{w(n)\}$ are uncorrelated.

- Determine the autocorrelation sequences $\{\gamma_{ss}(m)\}$ and $\{\gamma_{sx}(m)\}$.
 - Design a Wiener filter of length $M = 2$ to estimate $\{s(n)\}$.
 - Determine the MMSE for $M = 2$.
- 11.25** Determine the optimum causal IIR Wiener filter for the signal given in Problem 11.24 and the corresponding MMSE_∞ .
- 11.26** Determine the system function for the noncausal IIR Wiener filter for the signal given in Problem 11.24 and the corresponding MMSE_∞ .
- 11.27** Determine the optimum FIR Wiener filter of length $M = 3$ for the signal in Example 11.6.1 and the corresponding MMSE_3 . Compare MMSE_3 with MMSE_2 and comment on the difference.
- 11.28** An AR(2) process is defined by the difference equation

$$x(n) = x(n-1) - 0.6x(n-2) + w(n)$$

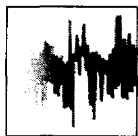
where $\{w(n)\}$ is a white noise process with variance σ_w^2 . Use the Yule-Walker equations to solve for the values of the autocorrelation $\gamma_{xx}(0)$, $\gamma_{xx}(1)$, and $\gamma_{xx}(2)$.

- 11.29** An observed random process $\{x(n)\}$ consists of the sum of an AR(p) process of the form

$$s(n) = - \sum_{k=1}^p a_p(k)s(n-k) + v(n)$$

and a white noise process $\{w(n)\}$ with variance σ_w^2 . The random process $\{v(n)\}$ is also white with variance σ_v^2 . The sequences $\{v(n)\}$ and $\{w(n)\}$ are uncorrelated.

Show that the observed process $\{x(n) = s(n) + w(n)\}$ is ARMA(p, p) and determine the coefficients of the numerator polynomial (MA component) in the corresponding system function.



12

Power Spectrum Estimation

In this chapter we are concerned with the estimation of the spectral characteristics of signals characterized as random processes. Many of the phenomena that occur in nature are best characterized statistically in terms of averages. For example, meteorological phenomena such as the fluctuations in air temperature and pressure are best characterized statistically as random processes. Thermal noise voltages generated in resistors and electronic devices are additional examples of physical signals that are well modeled as random processes.

Due to the random fluctuations in such signals, we must adopt a statistical viewpoint, which deals with the average characteristics of random signals. In particular, the autocorrelation function of a random process is the appropriate statistical average that we will use for characterizing random signals in the time domain, and the Fourier transform of the autocorrelation function, which yields the power density spectrum, provides the transformation from the time domain to the frequency domain.

Power spectrum estimation methods have a relatively long history. For a historical perspective, the reader is referred to the paper by Robinson (1982) and the book by Marple (1987). Our treatment of this subject covers the classical power spectrum estimation methods based on the periodogram, originally introduced by Schuster (1898), and by Yule (1927), who originated the modern model-based or parametric methods. These methods were subsequently developed and applied by Walker (1931), Bartlett (1948), Parzen (1957), Blackman and Tukey (1958), Burg (1967), and others. We also describe the method of Capon (1969) and methods based on eigenanalysis of the data correlation matrix.

12.1 ESTIMATION OF SPECTRA FROM FINITE-DURATION OBSERVATIONS OF SIGNALS

The basic problem that we consider in this chapter is the estimation of the power density spectrum of a signal from the observation of the signal over a finite time interval. As we will see, the finite record length of the data sequence is a major

limitation on the quality of the power spectrum estimate. When dealing with signals that are statistically stationary, the longer the data record, the better the estimate that can be extracted from the data. On the other hand, if the signal statistics are nonstationary, we cannot select an arbitrarily long data record to estimate the spectrum. In such a case, the length of the data record that we select is determined by the rapidity of the time variations in the signal statistics. Ultimately, our goal is to select as short a data record as possible that still allows us to resolve the spectral characteristics of different signal components in the data record that have closely spaced spectra.

One of the problems that we encounter with classical power spectrum estimation methods based on a finite-length data record is the distortion of the spectrum that we are attempting to estimate. This problem occurs in both the computation of the spectrum for a deterministic signal and the estimation of the power spectrum of a random signal. Since it is easier to observe the effect of the finite length of the data record on a deterministic signal, we treat this case first. Thereafter, we consider only random signals and the estimation of their power spectra.

12.1.1 Computation of the Energy Density Spectrum

Let us consider the computation of the spectrum of a deterministic signal from a finite sequence of data. The sequence $x(n)$ is usually the result of sampling a continuous-time signal $x_a(t)$ at some uniform sampling rate F_s . Our objective is to obtain an estimate of the true spectrum from a finite-duration sequence $x(n)$.

Recall that if $x(t)$ is a finite-energy signal, that is,

$$E = \int_{-\infty}^{\infty} |x_a(t)|^2 dt < \infty$$

then its Fourier transform exists and is given as

$$X_a(F) = \int_{-\infty}^{\infty} x_a(t) e^{-j2\pi Ft} dt$$

From Parseval's theorem we have

$$E = \int_{-\infty}^{\infty} |x_a(t)|^2 dt = \int_{-\infty}^{\infty} |X_a(F)|^2 dF \quad (12.1.1)$$

The quantity $|X_a(F)|^2$ represents the distribution of signal energy as a function of frequency, and hence it is called the energy density spectrum of the signal, that is,

$$S_{xx}(F) = |X_a(F)|^2 \quad (12.1.2)$$

as described in Chapter 4. Thus the total energy in the signal is simply the integral of $S_{xx}(F)$ over all F [i.e., the total area under $S_{xx}(F)$].

It is also interesting to note that $S_{xx}(F)$ can be viewed as the Fourier transform of another function, $R_{xx}(\tau)$, called the *autocorrelation function* of the

finite-energy signal $x_a(t)$, defined as

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x_a^*(t) x_a(t + \tau) dt \quad (12.1.3)$$

Indeed, it easily follows that

$$\int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j2\pi F\tau} d\tau = S_{xx}(F) = |X_a(F)|^2 \quad (12.1.4)$$

so that $R_{xx}(\tau)$ and $S_{xx}(F)$ are a Fourier transform pair.

Now suppose that we compute the energy density spectrum of the signal $x_a(t)$ from its samples taken at the rate F_s samples per second. To ensure that there is no spectral aliasing resulting from the sampling process, the signal is assumed to be prefiltered, so that, for practical purposes, its bandwidth is limited to B hertz. Then the sampling frequency F_s is selected such that $F_s > 2B$.

The sampled version of $x_a(t)$ is a sequence $x(n)$, $-\infty < n < \infty$, which has a Fourier transform (voltage spectrum)

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

or, equivalently,

$$X(f) = \sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi f n} \quad (12.1.5)$$

Recall that $X(f)$ can be expressed in terms of the voltage spectrum of the analog signal $x_a(t)$ as

$$X\left(\frac{F}{F_s}\right) = F_s \sum_{k=-\infty}^{\infty} X_a(F - kF_s) \quad (12.1.6)$$

where $f = F/F_s$ is the normalized frequency variable.

In the absence of aliasing, within the fundamental range $|F| \leq F_s/2$, we have

$$X\left(\frac{F}{F_s}\right) = F_s X_a(F), \quad |F| \leq F_s/2 \quad (12.1.7)$$

Hence the voltage spectrum of the sampled signal is identical to the voltage spectrum of the analog signal. As a consequence, the energy density spectrum of the sampled signal is

$$S_{xx}\left(\frac{F}{F_s}\right) = \left|X\left(\frac{F}{F_s}\right)\right|^2 = F_s^2 |X_a(F)|^2 \quad (12.1.8)$$

We can proceed further by noting that the autocorrelation of the sampled signal, which is defined as

$$r_{xx}(k) = \sum_{n=-\infty}^{\infty} x^*(n) x(n+k) \quad (12.1.9)$$

has the Fourier transform (Wiener-Khinchine theorem)

$$S_{xx}(f) = \sum_{k=-\infty}^{\infty} r_{xx}(k) e^{-j2\pi kf} \quad (12.1.10)$$

Hence the energy density spectrum can be obtained by the Fourier transform of the autocorrelation of the sequence $\{x(n)\}$.

The relations above lead us to distinguish between two distinct methods for computing the energy density spectrum of a signal $x_a(t)$ from its samples $x(n)$. One is the *direct method*, which involves computing the Fourier transform of $\{x(n)\}$, and then

$$\begin{aligned} S_{xx}(f) &= |X(f)|^2 \\ &= \left| \sum_{n=-\infty}^{\infty} x(n)e^{-j2\pi fn} \right|^2 \end{aligned} \quad (12.1.11)$$

The second approach is called the *indirect method* because it requires two steps. First, the autocorrelation $r_{xx}(k)$ is computed from $x(n)$ and then the Fourier transform of the autocorrelation is computed as in (12.1.10) to obtain the energy density spectrum.

In practice, however, only the finite-duration sequence $x(n)$, $0 \leq n \leq N-1$, is available for computing the spectrum of the signal. In effect, limiting the duration of the sequence $x(n)$ to N points is equivalent to multiplying $x(n)$ by a rectangular window. Thus we have

$$\tilde{x}(n) = x(n)w(n) = \begin{cases} x(n), & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases} \quad (12.1.12)$$

From our discussion of FIR filter design based on the use of windows to limit the duration of the impulse response, we recall that multiplication of two sequences is equivalent to convolution of their voltage spectra. Consequently, the frequency-domain relation corresponding to (12.1.12) is

$$\begin{aligned} \tilde{X}(f) &= X(f) * W(f) \\ &= \int_{-1/2}^{1/2} X(\alpha)W(f-\alpha)d\alpha \end{aligned} \quad (12.1.13)$$

Recall from our discussion in Section 8.2.1 that convolution of the window function $W(f)$ with $X(f)$ smooths the spectrum $\tilde{X}(f)$, provided that the spectrum $W(f)$ is relatively narrow compared to $X(f)$. But this condition implies that the window $w(n)$ be sufficiently long (i.e., N must be sufficiently large) such that $W(f)$ is narrow compared to $X(f)$. Even if $W(f)$ is narrow compared to $X(f)$, the convolution of $X(f)$ with the sidelobes of $W(f)$ results in sidelobe energy in $\tilde{X}(f)$, in frequency bands where the true signal spectrum $X(f) = 0$. This sidelobe energy is called *leakage*. The following example illustrates the leakage problem.

Example 12.1.1

A signal with (voltage) spectrum

$$X(f) = \begin{cases} 1, & |f| \leq 0.1 \\ 0, & \text{otherwise} \end{cases}$$

is convolved with the rectangular window of length $N = 61$. Determine the spectrum of $\tilde{X}(f)$ given by (12.1.13).

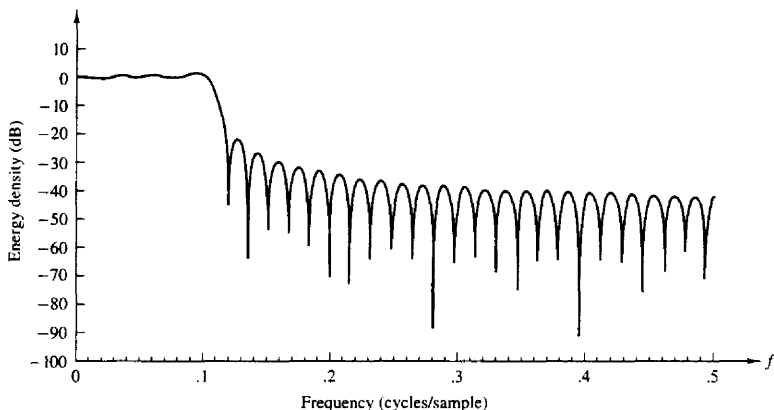


Figure 12.1 Spectrum obtained by convolving an $M = 61$ rectangular window with the ideal lowpass spectrum in Example 12.1.1.

Solution The spectral characteristic $W(f)$ for the length $N = 61$ rectangular window is illustrated in Fig. 8.2(b). Note that the width of the main lobe of the window function is $\Delta\omega = 4\pi/61$ or $\Delta f = 2/61$, which is narrow compared to $X(f)$.

The convolution of $X(f)$ with $W(f)$ is illustrated in Fig. 12.1. We note that energy has leaked into the frequency band $0.1 < |f| \leq 0.5$, where $X(f) = 0$. A part of this is due to the width of the main lobe in $W(f)$, which causes a broadening or smearing of $X(f)$ outside the range $|f| \leq 0.1$. However, the sidelobe energy in $\hat{X}(f)$ is due to the presence of the sidelobes of $W(f)$, which are convolved with $X(f)$. The smearing of $X(f)$ for $|f| > 0.1$ and the sidelobes in the range $0.1 \leq |f| \leq 0.5$ constitute the leakage.

Just as in the case of FIR filter design, we can reduce sidelobe leakage by selecting windows that have low sidelobes. This implies that the windows have a smooth time-domain cutoff instead of the abrupt cutoff in the rectangular window. Although such window functions reduce sidelobe leakage, they result in an increase in smoothing or broadening of the spectral characteristic $X(f)$. For example, the use of a Blackman window of length $N = 61$ in Example 12.1.1 results in the spectral characteristic $\hat{X}(f)$ shown in Fig. 12.2. The sidelobe leakage has certainly been reduced, but the spectral width has been increased by about 50%.

The broadening of the spectrum being estimated due to windowing is particularly a problem when we wish to resolve signals with closely spaced frequency components. For example, the signal with spectral characteristic $X(f) = X_1(f) + X_2(f)$, as shown in Fig. 12.3, cannot be resolved as two separate signals unless the width of the window function is significantly narrower than the frequency separation Δf . Thus we observe that using smooth time-domain windows reduces leakage at the expense of a decrease in frequency resolution.

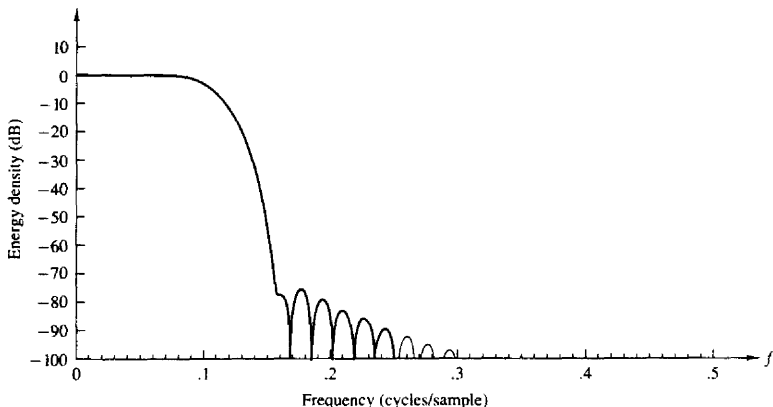


Figure 12.2 Spectrum obtained by convolving an $M = 61$ Blackman window with the ideal lowpass spectrum in Example 12.1.1.

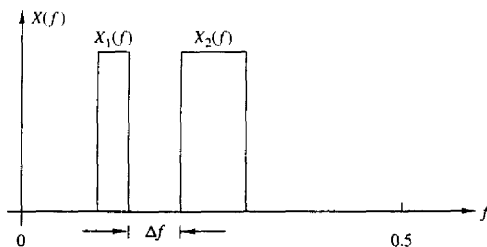


Figure 12.3 Two narrowband signal spectra.

It is clear from this discussion that the energy density spectrum of the windowed sequence $\{x(n)\}$ is an approximation of the desired spectrum of the sequence $\{x(n)\}$. The spectral density obtained from $\{\tilde{x}(n)\}$ is

$$S_{\tilde{x}\tilde{x}}(f) = |\tilde{X}(f)|^2 = \left| \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi f n} \right|^2 \quad (12.1.14)$$

The spectrum given by (12.1.14) can be computed numerically at a set of N frequency points by means of the DFT. Thus

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi k n / N} \quad (12.1.15)$$

Then

$$|\tilde{X}(k)|^2 = S_{\tilde{x}\tilde{x}}(f)|_{f=k/N} = S_{\tilde{x}\tilde{x}}\left(\frac{k}{N}\right) \quad (12.1.16)$$

and hence

$$S_{\tilde{x}\tilde{x}}\left(\frac{k}{N}\right) = \left| \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j2\pi kn/N} \right|^2 \quad (12.1.17)$$

which is a distorted version of the true spectrum $S_{xx}(k/N)$.

12.1.2 Estimation of the Autocorrelation and Power Spectrum of Random Signals: The Periodogram

The finite-energy signals considered in the preceding section possess a Fourier transform and are characterized in the spectral domain by their energy density spectrum. On the other hand, the important class of signals characterized as stationary random processes do not have finite energy and hence do not possess a Fourier transform. Such signals have finite average power and hence are characterized by a *power density spectrum*. If $x(t)$ is a stationary random process, its autocorrelation function is

$$\gamma_{xx}(\tau) = E[x^*(t)x(t+\tau)] \quad (12.1.18)$$

where $E[\cdot]$ denotes the statistical average. Then, via the Wiener–Khinchine theorem, the power density spectrum of the stationary random process is the Fourier transform of the autocorrelation function, that is,

$$\Gamma_{xx}(F) = \int_{-\infty}^{\infty} \gamma_{xx}(\tau) e^{-j2\pi F\tau} d\tau \quad (12.1.19)$$

In practice, we deal with a single realization of the random process from which we estimate the power spectrum of the process. We do not know the true autocorrelation function $\gamma_{xx}(\tau)$ and as a consequence, we cannot compute the Fourier transform in (12.1.19) to obtain $\Gamma_{xx}(F)$. On the other hand, from a single realization of the random process we can compute the time-average autocorrelation function

$$R_{xx}(\tau) = \frac{1}{2T_0} \int_{-T_0}^{T_0} x^*(t)x(t+\tau) dt \quad (12.1.20)$$

where $2T_0$ is the observation interval. If the stationary random process is *ergodic* in the first and second moments (mean and autocorrelation function), then

$$\begin{aligned} \gamma_{xx}(\tau) &= \lim_{T_0 \rightarrow \infty} R_{xx}(\tau) \\ &= \lim_{T_0 \rightarrow \infty} \frac{1}{2T_0} \int_{-T_0}^{T_0} x^*(t)x(t+\tau) dt \end{aligned} \quad (12.1.21)$$

This relation justifies the use of the time-average autocorrelation $R_{xx}(\tau)$ as an estimate of the statistical autocorrelation function $\gamma_{xx}(\tau)$. Furthermore, the Fourier transform of $R_{xx}(\tau)$ provides an estimate $P_{xx}(F)$ of the power density

spectrum, that is,

$$\begin{aligned}
 P_{xx}(F) &= \int_{-T_0}^{T_0} R_{xx}(\tau) e^{-j2\pi F\tau} d\tau \\
 &= \frac{1}{2T_0} \int_{-T_0}^{T_0} \left[\int_{-T_0}^{T_0} x^*(t)x(t+\tau) dt \right] e^{-j2\pi F\tau} d\tau \quad (12.1.22) \\
 &= \frac{1}{2T_0} \left| \int_{-T_0}^{T_0} x(t) e^{-j2\pi Ft} dt \right|^2
 \end{aligned}$$

The actual power density spectrum is the expected value of $P_{xx}(F)$ in the limit as $T_0 \rightarrow \infty$,

$$\begin{aligned}
 \Gamma_{xx}(F) &= \lim_{T_0 \rightarrow \infty} E[P_{xx}(F)] \\
 &= \lim_{T_0 \rightarrow \infty} E \left[\frac{1}{2T_0} \left| \int_{-T_0}^{T_0} x(t) e^{-j2\pi Ft} dt \right|^2 \right] \quad (12.1.23)
 \end{aligned}$$

From (12.1.20) and (12.1.22) we again note the two possible approaches to computing $P_{xx}(F)$, the direct method as given by (12.1.22) or the indirect method, in which we obtain $R_{xx}(\tau)$ first and then compute the Fourier transform.

We shall consider the estimation of the power density spectrum from samples of a single realization of the random process. In particular, we assume that $x_a(t)$ is sampled at a rate $F_s > 2B$, where B is the highest frequency contained in the power density spectrum of the random process. Thus we obtain a finite-duration sequence $x(n)$, $0 \leq n \leq N-1$, by sampling $x_a(t)$. From these samples we can compute the time-average autocorrelation sequence

$$\begin{aligned}
 r'_{xx}(m) &= \frac{1}{N-m} \sum_{n=0}^{N-m-1} x^*(n)x(n+m) \quad m = 0, 1, \dots, N-1 \\
 r'_{xx}(m) &= \frac{1}{N-|m|} \sum_{n=|m|}^{N-1} x^*(n)x(n+m) \quad m = -1, -2, \dots, 1-N
 \end{aligned} \quad (12.1.24)$$

and then compute the Fourier transform

$$P'_{xx}(f) = \sum_{m=-N+1}^{N-1} r'_{xx}(m) e^{-j2\pi f m} \quad (12.1.25)$$

The normalization factor $N-|m|$ in (12.1.24) results in an estimate with mean value

$$\begin{aligned}
 E[r'_{xx}(m)] &= \frac{1}{N-|m|} \sum_{n=0}^{N-m-1} E[x^*(n)x(n+m)] \\
 &= \gamma_{xx}(m)
 \end{aligned} \quad (12.1.26)$$

where $\gamma_{xx}(m)$ is the true (statistical) autocorrelation sequence of $x(n)$. Hence $r'_{xx}(m)$ is an unbiased estimate of the autocorrelation function $\gamma_{xx}(m)$. The variance

of the estimate $r'_{xx}(m)$ is approximately

$$\text{var}[r'_{xx}(m)] \approx \frac{N}{[N - |m|]^2} \sum_{n=-\infty}^{\infty} [|\gamma_{xx}(n)|^2 + \gamma_{xx}^*(n-m)\gamma_{xx}(n+m)] \quad (12.1.27)$$

which is a result given by Jenkins and Watts (1968). Clearly,

$$\lim_{N \rightarrow \infty} \text{var}[r'_{xx}(m)] = 0 \quad (12.1.28)$$

provided that

$$\sum_{n=-\infty}^{\infty} |\gamma_{xx}(n)|^2 < \infty$$

Since $E[r'_{xx}(m)] = \gamma_{xx}(m)$ and the variance of the estimate converges to zero as $N \rightarrow \infty$, the estimate $r'_{xx}(m)$ is said to be *consistent*.

For large values of the lag parameter m , the estimate $r'_{xx}(m)$ given by (12.1.24) has a large variance, especially as m approaches N . This is due to the fact that fewer data points enter into the estimate for large lags. As an alternative to (12.1.24) we can use the estimate

$$\begin{aligned} r_{xx}(m) &= \frac{1}{N} \sum_{n=0}^{N-m-1} x^*(n)x(n+m) \quad 0 \leq m \leq N-1 \\ r_{xx}(m) &= \frac{1}{N} \sum_{n=|m|}^{N-1} x^*(n)x(n+m) \quad m = -1, -2, \dots, 1-N \end{aligned} \quad (12.1.29)$$

which has a bias of $|m|\gamma_{xx}(m)/N$, since its mean value is

$$\begin{aligned} E[r_{xx}(m)] &= \frac{1}{N} \sum_{n=0}^{N-m-1} E[x^*(n)x(n+m)] \\ &= \frac{N - |m|}{N} \gamma_{xx}(m) = \left(1 - \frac{|m|}{N}\right) \gamma_{xx}(m) \end{aligned} \quad (12.1.30)$$

However, this estimate has a smaller variance, given approximately as

$$\text{var}[r_{xx}(m)] \approx \frac{1}{N} \sum_{n=-\infty}^{\infty} [\gamma_{xx}(n)|^2 + \gamma_{xx}^*(n-m)\gamma_{xx}(n+m)] \quad (12.1.31)$$

We observe that $r_{xx}(m)$ is *asymptotically unbiased*, that is,

$$\lim_{N \rightarrow \infty} E[r_{xx}(m)] = \gamma_{xx}(m) \quad (12.1.32)$$

and its variance converges to zero as $N \rightarrow \infty$. Therefore, the estimate $r_{xx}(m)$ is also a *consistent estimate* of $\gamma_{xx}(m)$.

We shall use the estimate $r_{xx}(m)$ given by (12.1.29) in our treatment of power spectrum estimation. The corresponding estimate of the power density spectrum is

$$P_{xx}(f) = \sum_{m=-(N-1)}^{N-1} r_{xx}(m) e^{-j2\pi fm} \quad (12.1.33)$$

If we substitute for $r_{xx}(m)$ from (12.1.29) into (12.1.33), the estimate $P_{xx}(f)$ can also be expressed as

$$P_{xx}(f) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x(n) e^{-j2\pi f n} \right|^2 = \frac{1}{N} |X(f)|^2 \quad (12.1.34)$$

where $X(f)$ is the Fourier transform of the sample sequence $x(n)$. This well known form of the power density spectrum estimate is called the *periodogram*. It was originally introduced by Schuster (1898) to detect and measure “hidden periodicities” in data.

From (12.1.33), the average value of the periodogram estimate $P_{xx}(f)$ is

$$\begin{aligned} E[P_{xx}(f)] &= E \left[\sum_{m=-(N-1)}^{N-1} r_{xx}(m) e^{-j2\pi f m} \right] = \sum_{m=-(N-1)}^{N-1} E[r_{xx}(m)] e^{-j2\pi f m} \\ E[P_{xx}(f)] &= \sum_{m=-(N-1)}^{N-1} \left(1 - \frac{|m|}{N} \right) \gamma_{xx}(m) e^{-j2\pi f m} \end{aligned} \quad (12.1.35)$$

The interpretation that we give to (12.1.35) is that the mean of the estimated spectrum is the Fourier transform of the windowed autocorrelation function

$$\tilde{\gamma}_{xx}(m) = \left(1 - \frac{|m|}{N} \right) \gamma_{xx}(m) \quad (12.1.36)$$

where the window function is the (triangular) Bartlett window. Hence the mean of the estimated spectrum is

$$\begin{aligned} E[P_{xx}(f)] &= \sum_{m=-\infty}^{\infty} \tilde{\gamma}_{xx}(m) e^{-j2\pi f m} \\ &= \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) W_B(f - \alpha) d\alpha \end{aligned} \quad (12.1.37)$$

where $W_B(f)$ is the spectral characteristic of the Bartlett window. The relation (12.1.37) illustrates that the mean of the estimated spectrum is the convolution of the true power density spectrum $\Gamma_{xx}(f)$ with the Fourier transform $W_B(f)$ of the Bartlett window. Consequently, the mean of the estimated spectrum is a smoothed version of the true spectrum and suffers from the same spectral leakage problems which are due to the finite number of data points.

We observe that the estimated spectrum is asymptotically unbiased, that is,

$$\lim_{N \rightarrow \infty} E \left[\sum_{m=-(N-1)}^{N-1} r_{xx}(m) e^{-j2\pi f m} \right] = \sum_{m=-\infty}^{\infty} \gamma_{xx}(m) e^{-j2\pi f m} = \Gamma_{xx}(f)$$

However, in general, the variance of the estimate $P_{xx}(f)$ does not decay to zero as $N \rightarrow \infty$. For example, when the data sequence is a Gaussian random process,

the variance is easily shown to be (see Problem 12.4)

$$\text{var}[P_{xx}(f)] = \Gamma_{xx}^2(f) \left[1 + \left(\frac{\sin 2\pi f N}{N \sin 2\pi f} \right)^2 \right] \quad (12.1.38)$$

which, in the limit as $N \rightarrow \infty$, becomes

$$\lim_{N \rightarrow \infty} \text{var}[P_{xx}(f)] = \Gamma_{xx}^2(f) \quad (12.1.39)$$

Hence we conclude that the *periodogram* is *not a consistent estimate of the true power density spectrum* (i.e., it does not converge to the true power density spectrum).

In summary, the estimated autocorrelation $r_{xx}(m)$ is a consistent estimate of the true autocorrelation function $\gamma_{xx}(m)$. However, its Fourier transform $P_{xx}(f)$, the periodogram, is not a consistent estimate of the true power density spectrum. We observed that $P_{xx}(f)$ is an asymptotically unbiased estimate of $\Gamma_{xx}(f)$, but for a finite-duration sequence, the mean value of $P_{xx}(f)$ contains a bias, which from (12.1.37) is evident as a distortion of the true power density spectrum. Thus the estimated spectrum suffers from the smoothing effects and the leakage embodied in the Bartlett window. The smoothing and leakage ultimately limit our ability to resolve closely spaced spectra.

The problems of leakage and frequency resolution that we have just described as well as the problem that the periodogram is not a consistent estimate of the power spectrum, provide the motivation for the power spectrum estimation methods described in Sections 12.2, 12.3, and 12.4. The methods described in Section 12.2 are classical nonparametric methods, which make no assumptions about the data sequence. The emphasis of the classical methods is on obtaining a consistent estimate of the power spectrum through some averaging or smoothing operations performed directly on the periodogram or on the autocorrelation. As we will see, the effect of these operations is to reduce the frequency resolution further, while the variance of the estimate is decreased.

The spectrum estimation methods described in Section 12.3 are based on some model of how the data were generated. In general, the model-based methods that have been developed over the past two decades provide significantly higher resolution than do the classical methods.

Additional methods are described in Sections 12.4 and 12.5. One of these methods, due to Capon (1969), is based on minimizing the variance in the spectral estimate. The methods described in Section 12.5 are based on an eigenvalue/eigenvector decomposition of the data correlation matrix.

12.1.3 The Use of the DFT in Power Spectrum Estimation

As given by (12.1.14) and (12.1.34), the estimated energy density spectrum $S_{xx}(f)$ and the periodogram $P_{xx}(f)$, respectively, can be computed by use of the DFT, which in turn is efficiently computed by a FFT algorithm. If we have N data points,

we compute as a minimum the N -point DFT. For example, the computation yields samples of the periodogram

$$P_{xx}\left(\frac{k}{N}\right) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/N} \right|^2 \quad k = 0, 1, \dots, N-1 \quad (12.1.40)$$

at the frequencies $f_k = k/N$.

In practice, however, such a sparse sampling of the spectrum does not provide a very good representation or a good picture of the continuous spectrum estimate $P_{xx}(f)$. This is easily remedied by evaluating $P_{xx}(f)$ at additional frequencies. Equivalently, we can effectively increase the length of the sequence by means of zero padding and then evaluate $P_{xx}(f)$ at a more dense set of frequencies. Thus if we increase the data sequence length to L points by means of zero padding and evaluate the L -point DFT, we have

$$P_{xx}\left(\frac{k}{L}\right) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x(n) e^{-j2\pi nk/L} \right|^2 \quad k = 0, 1, \dots, L-1 \quad (12.1.41)$$

We emphasize that zero padding and evaluating the DFT at $L > N$ points does not improve the frequency resolution in the spectral estimate. It simply provides us with a method for interpolating the values of the measured spectrum at more frequencies. The frequency resolution in the spectral estimate $P_{xx}(f)$ is determined by the length N of the data record.

Example 12.1.2

A sequence of $N = 16$ samples is obtained by sampling an analog signal consisting of two frequency components. The resulting discrete-time sequence is

$$x(n) = \sin 2\pi(0.135)n + \cos 2\pi(0.135 + \Delta f)n \quad n = 0, 1, \dots, 15$$

where Δf is the frequency separation. Evaluate the power spectrum $P(f) = (1/N)|X(f)|^2$ at the frequencies $f_k = k/L$, $k = 0, 1, \dots, L-1$, for $L = 8, 16, 32$, and 128 for values of $\Delta f = 0.06$ and $\Delta f = 0.01$.

Solution By zero padding, we increase the data sequence to obtain the power spectrum estimate $P_{xx}(k/L)$. The results for $\Delta f = 0.06$ are plotted in Fig. 12.4. Note that zero padding does not change the resolution, but it does have the effect of interpolating the spectrum $P_{xx}(f)$. In this case the frequency separation Δf is sufficiently large so that the two frequency components are resolvable.

The spectral estimates for $\Delta f = 0.01$ are shown in Fig. 12.5. In this case the two spectral components are not resolvable. Again, the effect of zero padding is to provide more interpolation, thus giving us a better picture of the estimated spectrum. It does not improve the frequency resolution.

When only a few points of the periodogram are needed, the Goertzel algorithm described in Chapter 6 may provide a more efficient computation. Since the Goertzel algorithm has been interpreted as a linear filtering approach to computing the DFT, it is clear that the periodogram estimate can be obtained by passing

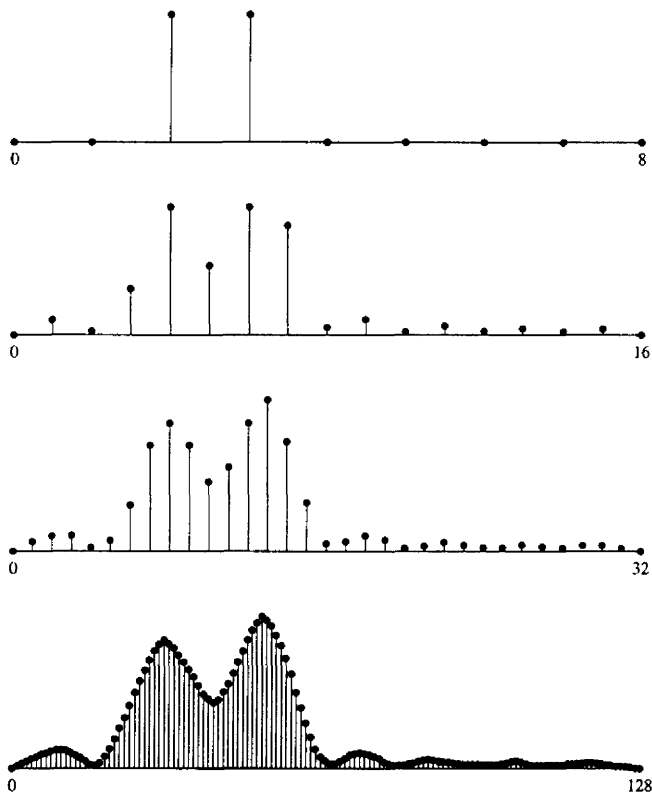


Figure 12.4 Spectra of two sinusoids with frequency separation $\Delta f = 0.06$.

the signal through a bank of parallel tuned filters and squaring their outputs (see Problem 12.5).

12.2 NONPARAMETRIC METHODS FOR POWER SPECTRUM ESTIMATION

The power spectrum estimation methods described in this section are the classical methods developed by Bartlett (1948), Blackman and Tukey (1958), and Welch (1967). These methods make no assumption about how the data were generated and hence are called *nonparametric*.

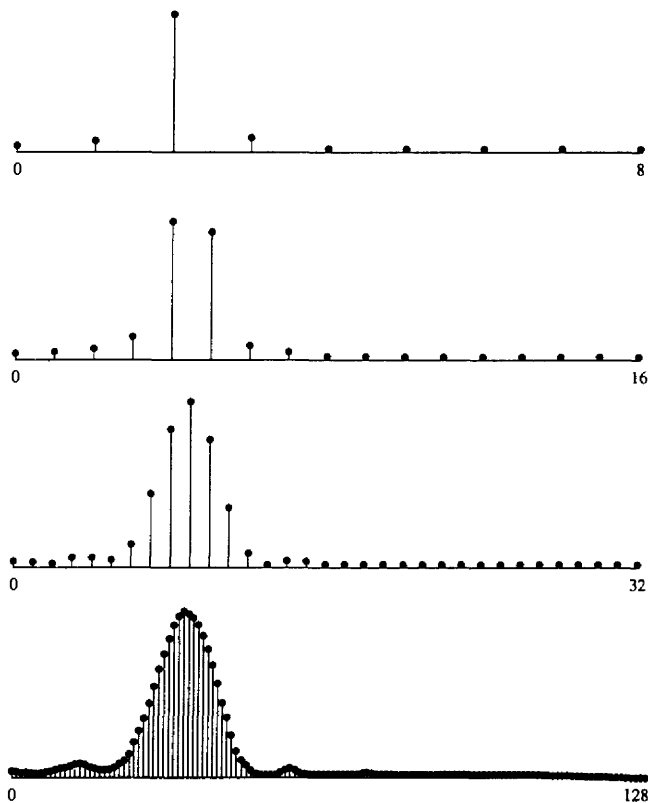


Figure 12.5 Spectra of two sinusoids with frequency separation $\Delta f = 0.01$.

Since the estimates are based entirely on a finite record of data, the frequency resolution of these methods is, at best, equal to the spectral width of the rectangular window of length N , which is approximately $1/N$ at the -3 -dB points. We shall be more precise in specifying the frequency resolution of the specific methods. All the estimation techniques described in this section decrease the frequency resolution in order to reduce the variance in the spectral estimate.

First, we describe the estimates and derive the mean and variance of each. A comparison of the three methods is given in Section 12.2.4. Although the spectral estimates are expressed as a function of the continuous frequency variable f , in practice, the estimates are computed at discrete frequencies via the FFT algorithm. The FFT-based computational requirements are considered in Section 12.2.5.

12.2.1 The Bartlett Method: Averaging Periodograms

Bartlett's method for reducing the variance in the periodogram involves three steps. First, the N -point sequence is subdivided into K nonoverlapping segments, where each segment has length M . This results in the K data segments

$$\begin{aligned} x_i(n) &= x(n + iM) & i &= 0, 1, \dots, K-1 \\ n &= 0, 1, \dots, M-1 \end{aligned} \quad (12.2.1)$$

For each segment, we compute the periodogram

$$P_{xx}^{(i)}(f) = \frac{1}{M} \left| \sum_{n=0}^{M-1} x_i(n) e^{-j2\pi fn} \right|^2 \quad i = 0, 1, \dots, K-1 \quad (12.2.2)$$

Finally, we average the periodograms for the K segments to obtain the Bartlett power spectrum estimate [Bartlett (1948)]

$$P_{xx}^B(f) = \frac{1}{K} \sum_{i=0}^{K-1} P_{xx}^{(i)}(f) \quad (12.2.3)$$

The statistical properties of this estimate are easily obtained. First, the mean value is

$$\begin{aligned} E[P_{xx}^B(f)] &= \frac{1}{K} \sum_{i=0}^{K-1} E[P_{xx}^{(i)}(f)] \\ &= E[P_{xx}^{(i)}(f)] \end{aligned} \quad (12.2.4)$$

From (12.1.35) and (12.1.37) we have the expected value for the single periodogram as

$$\begin{aligned} E[P_{xx}^{(i)}(f)] &= \sum_{m=-(M-1)}^{M-1} \left(1 - \frac{|m|}{M}\right) \gamma_{xx}(m) e^{-j2\pi fm} \\ &= \frac{1}{M} \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) \left(\frac{\sin \pi(f - \alpha)M}{\sin \pi(f - \alpha)} \right)^2 d\alpha \end{aligned} \quad (12.2.5)$$

where

$$W_B(f) = \frac{1}{M} \left(\frac{\sin \pi f M}{\sin \pi f} \right)^2 \quad (12.2.6)$$

is the frequency characteristics of the Bartlett window

$$w_B(n) = \begin{cases} 1 - \frac{|m|}{M}, & |m| \leq M-1 \\ 0, & \text{otherwise} \end{cases} \quad (12.2.7)$$

From (12.2.5) we observe that the true spectrum is now convolved with the frequency characteristic $W_B(f)$ of the Bartlett window. The effect of reducing the length of the data from N points to $M = N/K$ results in a window whose spectral

width has been increased by a factor of K . Consequently, the frequency resolution has been reduced by a factor K .

In return for this reduction in resolution, we have reduced the variance. The variance of the Bartlett estimate is

$$\begin{aligned}\text{var}[P_{xx}^B(f)] &= \frac{1}{K^2} \sum_{i=0}^{K-1} \text{var}[P_{xx}^{(i)}(f)] \\ &= \frac{1}{K} \text{var}[P_{xx}^{(i)}(f)]\end{aligned}\quad (12.2.8)$$

If we make use of (12.1.38) in (12.2.8), we obtain

$$\text{var}[P_{xx}^B(f)] = \frac{1}{K} \Gamma_{xx}^2(f) \left[1 + \left(\frac{\sin 2\pi f M}{M \sin 2\pi f} \right)^2 \right] \quad (12.2.9)$$

Therefore, the variance of the Bartlett power spectrum estimate has been reduced by the factor K .

12.2.2 The Welch Method: Averaging Modified Periodograms

Welch (1967) made two basic modifications to the Bartlett method. First, he allowed the data segments to overlap. Thus the data segments can be represented as

$$\begin{aligned}x_i(n) &= x(n + iD) & n = 0, 1, \dots, M-1 \\ & & i = 0, 1, \dots, L-1\end{aligned}\quad (12.2.10)$$

where iD is the starting point for the i th sequence. Observe that if $D = M$, the segments do not overlap and the number L of data segments is identical to the number K in the Bartlett method. However, if $D = M/2$, there is 50% overlap between successive data segments and $L = 2K$ segments are obtained. Alternatively, we can form K data segments each of length $2M$.

The second modification made by Welch to the Bartlett method is to window the data segments prior to computing the periodogram. The result is a "modified" periodogram

$$\tilde{P}_{xx}^{(i)}(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} x_i(n) w(n) e^{-j2\pi f n} \right|^2 \quad i = 0, 1, \dots, L-1 \quad (12.2.11)$$

where U is a normalization factor for the power in the window function and is selected as

$$U = \frac{1}{M} \sum_{n=0}^{M-1} w^2(n) \quad (12.2.12)$$

The Welch power spectrum estimate is the average of these modified periodograms, that is,

$$P_{xx}^W(f) = \frac{1}{L} \sum_{i=0}^{L-1} \tilde{P}_{xx}^{(i)}(f) \quad (12.2.13)$$

The mean value of the Welch estimate is

$$\begin{aligned} E[P_{xx}^W(f)] &= \frac{1}{L} \sum_{i=0}^{L-1} E[\tilde{P}_{xx}^{(i)}(f)] \\ &= E[\tilde{P}_{xx}^{(i)}(f)] \end{aligned} \quad (12.2.14)$$

But the expected value of the modified periodogram is

$$\begin{aligned} E[\tilde{P}_{xx}^{(i)}(f)] &= \frac{1}{MU} \sum_{n=0}^{M-1} \sum_{m=0}^{M-1} w(n)w(m) E[x_i(n)x_i^*(m)] e^{-j2\pi f(n-m)} \\ &= \frac{1}{MU} \sum_{n=0}^{M-1} \sum_{m=0}^{M-1} w(n)w(m) \gamma_{xx}(n-m) e^{-j2\pi f(n-m)} \end{aligned} \quad (12.2.15)$$

Since

$$\gamma_{xx}(n) = \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) e^{j2\pi \alpha n} d\alpha \quad (12.2.16)$$

substitution for $\gamma_{xx}(n)$ from (12.2.16) into (12.2.15) yields

$$\begin{aligned} E[\tilde{P}_{xx}^{(i)}(f)] &= \frac{1}{MU} \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) \left[\sum_{n=0}^{M-1} \sum_{m=0}^{M-1} w(n)w(m) e^{-j2\pi(n-m)(f-\alpha)} \right] d\alpha \\ &= \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha) W(f-\alpha) d\alpha \end{aligned} \quad (12.2.17)$$

where, by definition,

$$W(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} w(n) e^{-j2\pi f n} \right|^2 \quad (12.2.18)$$

The normalization factor U ensures that

$$\int_{-1/2}^{1/2} W(f) df = 1 \quad (12.2.19)$$

The variance of the Welch estimate is

$$\text{var}[P_{xx}^W(f)] = \frac{1}{L^2} \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} E[\tilde{P}_{xx}^{(i)}(f) \tilde{P}_{xx}^{(j)}(f)] - \{E[P_{xx}^W(f)]\}^2 \quad (12.2.20)$$

In the case of no overlap between successive data segments ($L = K$), Welch has shown that

$$\begin{aligned} \text{var}[P_{xx}^W(f)] &= \frac{1}{L} \text{var}[\tilde{P}_{xx}^{(i)}(f)] \\ &\approx \frac{1}{L} \Gamma_{xx}^2(f) \end{aligned} \quad (12.2.21)$$

In the case of 50% overlap between successive data segments ($L = 2K$), the variance of the Welch power spectrum estimate with the Bartlett (triangular) window, also derived in the paper by Welch, is

$$\text{var}[P_{xx}^W(f)] \approx \frac{9}{8L} \Gamma_{xx}^2(f) \quad (12.2.22)$$

Although we considered only the triangular window in the computation of the variance, other window functions may be used. In general, they will yield a different variance. In addition, one may vary the data segment overlapping by either more or less than the 50% considered in this section in an attempt to improve the relevant characteristics of the estimate.

12.2.3 The Blackman and Tukey Method: Smoothing the Periodogram

Blackman and Tukey (1958) proposed and analyzed the method in which the sample autocorrelation sequence is windowed first and then Fourier transformed to yield the estimate of the power spectrum. The rationale for windowing the estimated autocorrelation sequence $r_{xx}(m)$ is that, for large lags, the estimates are less reliable because a smaller number ($N - m$) of data points enter into the estimate. For values of m approaching N , the variance of these estimates is very high, and hence these estimates should be given a smaller weight in the formation of the estimated power spectrum. Thus the Blackman-Tukey estimate is

$$P_{xx}^{BT}(f) = \sum_{m=-(M-1)}^{M-1} r_{xx}(m)w(m)e^{-j2\pi fm} \quad (12.2.23)$$

where the window function $w(n)$ has length $2M - 1$ and is zero for $|m| \geq M$. With this definition for $w(n)$, the limits on the sum in (12.2.23) can be extended to $(-\infty, \infty)$. Hence the frequency-domain equivalent expression for (12.2.23) is the convolution integral

$$P_{xx}^{BT}(f) = \int_{-1/2}^{1/2} P_{xx}(\alpha)W(f - \alpha)d\alpha \quad (12.2.24)$$

where $P_{xx}(f)$ is the periodogram. It is clear from (12.2.24) that the effect of windowing the autocorrelation is to smooth the periodogram estimate, thus decreasing the variance in the estimate at the expense of reducing the resolution.

The window sequence $w(n)$ should be symmetric (even) about $m = 0$ to ensure that the estimate of the power spectrum is real. Furthermore, it is desirable to select the window spectrum to be nonnegative, that is,

$$W(f) \geq 0 \quad |f| \leq 1/2 \quad (12.2.25)$$

This condition ensures that $P_{xx}^{BT}(f) \geq 0$ for $|f| \leq 1/2$, which is a desirable property for any power spectrum estimate. We should indicate, however, that some of the window functions we have introduced do not satisfy this condition. For example,

in spite of their low sidelobe levels, the Hamming and Hann (or Hanning) windows do not satisfy the property in (12.2.25) and, consequently, may result in negative spectrum estimates in some parts of the frequency range.

The expected value of the Blackman-Tukey power spectrum estimate is

$$E[P_{xx}^{BT}(f)] = \int_{-1/2}^{1/2} E[P_{xx}(\alpha)] W(f - \alpha) d\alpha \quad (12.2.26)$$

where from (12.1.37) we have

$$E[P_{xx}(\alpha)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W_B(\alpha - \theta) d\theta \quad (12.2.27)$$

and $W_B(f)$ is the Fourier transform of the Bartlett window. Substitution of (12.2.27) into (12.2.26) yields the double convolution integral

$$E[P_{xx}^{BT}(f)] = \int_{-1/2}^{1/2} \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W_B(\alpha - \theta) W(f - \alpha) d\alpha d\theta \quad (12.2.28)$$

Equivalently, by working in the time domain, the expected value of the Blackman-Tukey power spectrum estimate is

$$\begin{aligned} E[P_{xx}^{BT}(f)] &= \sum_{m=-(M-1)}^{M-1} E[r_{xx}(m)] w(m) e^{-j2\pi fm} \\ &= \sum_{m=-(M-1)}^{M-1} \gamma_{xx}(m) w_B(m) w(m) e^{-j2\pi fm} \end{aligned} \quad (12.2.29)$$

where the Bartlett window is

$$w_B(m) = \begin{cases} 1 - \frac{|m|}{N}, & |m| < N \\ 0, & \text{otherwise} \end{cases} \quad (12.2.30)$$

Clearly, we should select the window length for $w(n)$ such that $M \ll N$, that is, $w(n)$ should be narrower than $w_B(m)$ to provide additional smoothing of the periodogram. Under this condition, (12.2.28) becomes

$$E[P_{xx}^{BT}(f)] \approx \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W(f - \theta) d\theta \quad (12.2.31)$$

since

$$\begin{aligned} \int_{-1/2}^{1/2} W_B(\alpha - \theta) W(f - \alpha) d\alpha &= \int_{-1/2}^{1/2} W_B(\alpha) W(f - \theta - \alpha) d\alpha \\ &\approx W(f - \theta) \end{aligned} \quad (12.2.32)$$

The variance of the Blackman-Tukey power spectrum estimate is

$$\text{var}[P_{xx}^{BT}(f)] = E\{[P_{xx}^{BT}(f)]^2\} - \{E[P_{xx}^{BT}(f)]\}^2 \quad (12.2.33)$$

where the mean can be approximated as in (12.2.31). The second moment in (11.2.33) is

$$E\{[P_{xx}^{BT}(f)]^2\} = \int_{-1/2}^{1/2} \int_{-1/2}^{1/2} E[P_{xx}(\alpha)P_{xx}(\theta)]W(f-\alpha)W(f-\theta)d\alpha d\theta \quad (12.2.34)$$

On the assumption that the random process is Gaussian (see Problem 12.5), we find that

$$E[P_{xx}(\alpha)P_{xx}(\theta)] = \Gamma_{xx}(\alpha)\Gamma_{xx}(\theta) \left\{ 1 + \left[\frac{\sin \pi(\theta + \alpha)N}{N \sin \pi(\theta + \alpha)} \right]^2 + \left[\frac{\sin \pi(\theta - \alpha)N}{N \sin \pi(\theta - \alpha)} \right]^2 \right\} \quad (12.2.35)$$

Substitution of (12.2.35) into (12.2.34) yields

$$\begin{aligned} E\{[P_{xx}^{BT}(f)]^2\} &= \left[\int_{-1/2}^{1/2} \Gamma_{xx}(\theta)W(f-\theta)d\theta \right]^2 \\ &+ \int_{-1/2}^{1/2} \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha)\Gamma_{xx}(\theta)W(f-\alpha)W(f-\theta) \\ &\times \left\{ \left[\frac{\sin \pi(\theta + \alpha)N}{N \sin \pi(\theta + \alpha)} \right]^2 + \left[\frac{\sin \pi(\theta - \alpha)N}{N \sin \pi(\theta - \alpha)} \right]^2 \right\} d\alpha d\theta \quad (12.2.36) \end{aligned}$$

The first term in (12.2.36) is simply the square of the mean of $P_{xx}^{BT}(f)$, which is to be subtracted out according to (12.2.33). This leaves the second term in (12.2.36), which constitutes the variance. For the case in which $N \gg M$, the functions $\sin \pi(\theta + \alpha)N/N \sin \pi(\theta + \alpha)$ and $\sin \pi(\theta - \alpha)N/N \sin \pi(\theta - \alpha)$ are relatively narrow compared to $W(f)$ in the vicinity of $\theta = -\alpha$ and $\theta = \alpha$, respectively. Therefore,

$$\begin{aligned} \int_{-1/2}^{1/2} \Gamma_{xx}(\theta)W(f-\theta) \left\{ \left[\frac{\sin \pi(\theta + \alpha)N}{N \sin \pi(\theta + \alpha)} \right]^2 + \left[\frac{\sin \pi(\theta - \alpha)N}{N \sin \pi(\theta - \alpha)} \right]^2 \right\} d\theta \\ \approx \frac{\Gamma_{xx}(-\alpha)W(f+\alpha) + \Gamma_{xx}(\alpha)W(f-\alpha)}{N} \quad (12.2.37) \end{aligned}$$

With this approximation, the variance of $P_{xx}^{BT}(f)$ becomes

$$\begin{aligned} \text{var}[P_{xx}^{BT}(f)] &\approx \frac{1}{N} \int_{-1/2}^{1/2} \Gamma_{xx}(\alpha)W(f-\alpha)[\Gamma_{xx}(-\alpha)W(f+\alpha) + \Gamma_{xx}(\alpha)W(f-\alpha)]d\alpha \\ &\approx \frac{1}{N} \int_{-1/2}^{1/2} \Gamma_{xx}^2(\alpha)W^2(f-\alpha)d\alpha \quad (12.2.38) \end{aligned}$$

where in the last step we made the approximation

$$\int_{-1/2}^{1/2} \Gamma_{xx}(\alpha)\Gamma_{xx}(-\alpha)W(f-\alpha)W(f+\alpha)d\alpha \approx 0 \quad (12.2.39)$$

We shall make one additional approximation in (12.2.38). When $W(f)$ is narrow compared to the true power spectrum $\Gamma_{xx}(f)$, (12.2.38) is further approximated as

$$\begin{aligned}\text{var}[P_{xx}^{BT}(f)] &\approx \Gamma_{xx}^2(f) \left[\frac{1}{N} \int_{-1/2}^{1/2} W^2(\theta) d\theta \right] \\ &\approx \Gamma_{xx}^2(f) \left[\frac{1}{N} \sum_{m=-(M-1)}^{M-1} w^2(m) \right]\end{aligned}\quad (12.2.40)$$

12.2.4 Performance Characteristics of Nonparametric Power Spectrum Estimators

In this section we compare the quality of the Bartlett, Welch, and Blackman and Tukey power spectrum estimates. As a measure of quality, we use the ratio of its variance to the square of the mean of the power spectrum estimate that is,

$$Q_A = \frac{(E[P_{xx}^A(f)])^2}{\text{var}[P_{xx}^A(f)]} \quad (12.2.41)$$

where $A = B, W$, or BT for the three power spectrum estimates. The reciprocal of this quantity, called the *variability*, can also be used as a measure of performance.

For reference, the periodogram has a mean and variance

$$E[P_{xx}(f)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W_B(f - \theta) d\theta \quad (12.2.42)$$

$$\text{var}[P_{xx}(f)] = \Gamma_{xx}^2(f) \left[1 + \left(\frac{\sin 2\pi f N}{N \sin 2\pi f} \right)^2 \right] \quad (12.2.43)$$

where

$$W_B(f) = \frac{1}{N} \left(\frac{\sin \pi f N}{\sin \pi f} \right)^2 \quad (12.2.44)$$

For large N (i.e., $N \rightarrow \infty$),

$$\begin{aligned}E[P_{xx}(f)] &\rightarrow \Gamma_{xx}(f) \int_{-1/2}^{1/2} W_B(\theta) d\theta = w_B(0) \Gamma_{xx}(f) = \Gamma_{xx}(f) \\ \text{var}[P_{xx}(f)] &\rightarrow \Gamma_{xx}^2(f)\end{aligned}\quad (12.2.45)$$

Hence, as indicated previously, the periodogram is an asymptotically unbiased estimate of the power spectrum, but it is not consistent because its variance does not approach zero as N increases toward infinity.

Asymptotically, the periodogram is characterized by the quality factor

$$Q_P = \frac{\Gamma_{xx}^2(f)}{\Gamma_{xx}^2(f)} = 1 \quad (12.2.46)$$

The fact that Q_P is fixed and independent of the data length N is another indication of the poor quality of this estimate.

Bartlett power spectrum estimate. The mean and variance of the Bartlett power spectrum estimate are

$$E[P_{xx}^B(f)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W_B(f - \theta) d\theta \quad (12.2.47)$$

$$\text{var}[P_{xx}^B(f)] = \frac{1}{K} \Gamma_{xx}^2(f) \left[1 + \left(\frac{\sin 2\pi f M}{M \sin 2\pi f} \right)^2 \right] \quad (12.2.48)$$

and

$$W_B(f) = \frac{1}{M} \left(\frac{\sin \pi f M}{\sin \pi f} \right)^2 \quad (12.2.49)$$

As $N \rightarrow \infty$ and $M \rightarrow \infty$, while $K = N/M$ remains fixed, we find that

$$E[P_{xx}^B(f)] \rightarrow \Gamma_{xx}(f) \int_{-1/2}^{1/2} W_B(f) df = \Gamma_{xx}(f) w_B(0) = \Gamma_{xx}(f) \quad (12.2.50)$$

$$\text{var}[P_{xx}^B(f)] \rightarrow \frac{1}{K} \Gamma_{xx}^2(f)$$

We observe that the Bartlett power spectrum estimate is asymptotically unbiased and if K is allowed to increase with an increase in N , the estimate is also consistent. Hence, asymptotically, this estimate is characterized by the quality factor

$$Q_B = K = \frac{N}{M} \quad (12.2.51)$$

The frequency resolution of the Bartlett estimate, measured by taking the 3-dB width of the main lobe of the rectangular window, is

$$\Delta f = \frac{0.9}{M} \quad (12.2.52)$$

Hence, $M = 0.9/\Delta f$ and the quality factor becomes

$$Q_B = \frac{N}{0.9/\Delta f} = 1.1N\Delta f \quad (12.2.53)$$

Welch power spectrum estimate. The mean and variance of the Welch power spectrum estimate are

$$E[P_{xx}^W(f)] = \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W(f - \theta) d\theta \quad (12.2.54)$$

where

$$W(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} w(n) e^{-j2\pi f n} \right|^2 \quad (12.2.55)$$

and

$$\text{var}[P_{xx}^W(f)] = \begin{cases} \frac{1}{L} \Gamma_{xx}^2(f), & \text{for no overlap} \\ \frac{9}{8L} \Gamma_{xx}^2(f), & \text{for 50\% overlap and triangular window} \end{cases} \quad (12.2.56)$$

As $N \rightarrow \infty$ and $M \rightarrow \infty$, the mean converges to

$$E[P_{xx}^w(f)] \rightarrow \Gamma_{xx}(f) \quad (12.2.57)$$

and the variance converges to zero, so that the estimate is consistent.

Under the two conditions given by (12.2.56) the quality factor is

$$Q_w = \begin{cases} L = \frac{N}{M}, & \text{for no overlap} \\ \frac{8L}{9} = \frac{16N}{9M}, & \text{for 50\% overlap and triangular window} \end{cases} \quad (12.2.58)$$

On the other hand, the spectral width of the triangular window at the 3-dB points is

$$\Delta f = \frac{1.28}{M} \quad (12.2.59)$$

Consequently, the quality factor expressed in terms of Δf and N is

$$Q_w = \begin{cases} 0.78N\Delta f, & \text{for no overlap} \\ 1.39N\Delta f, & \text{for 50\% overlap and triangular window} \end{cases} \quad (12.2.60)$$

Blackman-Tukey power spectrum estimate. The mean and variance of this estimate are approximated as

$$\begin{aligned} E[P_{xx}^{BT}(f)] &\approx \int_{-1/2}^{1/2} \Gamma_{xx}(\theta) W(f - \theta) d\theta \\ \text{var}[P_{xx}^{BT}(f)] &\approx \Gamma_{xx}^2(f) \left[\frac{1}{N} \sum_{m=-(M-1)}^{M-1} w^2(m) \right] \end{aligned} \quad (12.2.61)$$

where $w(m)$ is the window sequence used to taper the estimated autocorrelation sequence. For the rectangular and Bartlett (triangular) windows we have

$$\frac{1}{N} \sum_{n=-(M-1)}^{M-1} w^2(n) = \begin{cases} 2M/N, & \text{rectangular window} \\ 2M/3N, & \text{triangular window} \end{cases} \quad (12.2.62)$$

It is clear from (12.2.61) that the mean value of the estimate is asymptotically unbiased. Its quality factor for the triangular window is

$$Q_{BT} = 1.5 \frac{N}{M} \quad (12.2.63)$$

Since the window length is $2M - 1$, the frequency resolution measured at the 3-dB points is

$$\Delta f = \frac{1.28}{2M} = \frac{0.64}{M} \quad (12.2.64)$$

and hence

$$Q_{BT} = \frac{1.5}{0.64} N \Delta f = 2.34 N \Delta f \quad (12.2.65)$$

TABLE 12.1 QUALITY OF POWER SPECTRUM ESTIMATES

Estimate	Quality Factor
Bartlett	$1.11N\Delta f$
Welch (50% overlap)	$1.39N\Delta f$
Blackman–Tukey	$2.34N\Delta f$

These results are summarized in Table 12.1. It is apparent from the results we have obtained that the Welch and Blackman–Tukey power spectrum estimates are somewhat better than the Bartlett estimate. However, the differences in performance are relatively small. The main point is that the quality factor increases with an increase in the length N of the data. This characteristic behavior is not shared by the periodogram estimate. Furthermore, the quality factor depends on the product of the data length N and the frequency resolution Δf . For a desired level of quality, Δf can be decreased (frequency resolution increased) by increasing the length N of the data, and vice versa.

12.2.5 Computational Requirements of Nonparametric Power Spectrum Estimates

The other important aspect of the nonparametric power spectrum estimates is their computational requirements. For this comparison we assume the estimates are based on a fixed amount of data N and a specified resolution Δf . The radix-2 FFT algorithm is assumed in all the computations. We shall count only the number of complex multiplications required to compute the power spectrum estimate.

Bartlett power spectrum estimate

$$\text{FFT length} = M = 0.9/\Delta f$$

$$\text{Number of FFTs} = \frac{N}{M} = 1.11N\Delta f$$

$$\text{Number of computations} = \frac{N}{M} \left(\frac{M}{2} \log_2 M \right) = \frac{N}{2} \log_2 \frac{0.9}{\Delta f}$$

Welch power spectrum estimate (50% overlap)

$$\text{FFT length} = M = 1.28/\Delta f$$

$$\text{Number of FFTs} = \frac{2N}{M} = 1.56N\Delta f$$

$$\text{Number of computations} = \frac{2N}{M} \left(\frac{M}{2} \log_2 M \right) = N \log_2 \frac{1.28}{\Delta f}$$

In addition to the $2N/M$ FFTs, there are additional multiplications required for windowing the data. Each data record requires M multiplications. Therefore, the total number of computations is

$$\text{Total computations} = 2N + N \log_2 \frac{1.28}{\Delta f} = N \log_2 \frac{5.12}{\Delta f}$$

Blackman–Tukey power spectrum estimate. In the Blackman–Tukey method, the autocorrelation $r_{xx}(m)$ can be computed efficiently via the FFT algorithm. However, if the number of data points is large, it may not be possible to compute one N -point DFT. For example, we may have $N = 10^5$ data points but only the capacity to perform 1024-point DFTs. Since the autocorrelation sequence is windowed to $2M - 1$ points where $M \ll N$, it is possible to compute the desired $2M - 1$ points of $r_{xx}(m)$ by segmenting the data into $K = N/2M$ records and then computing $2M$ -point DFTs and one $2M$ -point IDFT via the FFT algorithm. Rader (1970) has described a method for performing this computation (see Problem 12.7).

If we base the computational complexity of the Blackman–Tukey method on this approach, we obtain the following computational requirements.

$$\text{FFT length} = 2M = 1.28/\Delta f$$

$$\text{Number of FFTs} = 2K + 1 = 2 \left(\frac{N}{2M} \right) + 1 \approx \frac{N}{M}$$

$$\text{Number of computations} = \frac{N}{M} (M \log_2 2M) = N \log_2 \frac{1.28}{\Delta f}$$

We can neglect the additional M multiplications required to window the autocorrelation sequence $r_{xx}(m)$, since it is a relatively small number. Finally, there is the additional computation required to perform the Fourier transform of the windowed autocorrelation sequence. The FFT algorithm can be used for this computation with some zero padding for purposes of interpolating the spectral estimate. As a result of these additional computations, the number of computations is increased by a small amount.

From these results we conclude that the Welch method requires a little more computational power than do the other two methods. The Bartlett method apparently requires the smallest number of computations. However, the differences in the computational requirements of the three methods are relatively small.

12.3 PARAMETRIC METHODS FOR POWER SPECTRUM ESTIMATION

The nonparametric power spectrum estimation methods described in the preceding section are relatively simple, well understood, and easy to compute using the FFT algorithm. However, these methods require the availability of long data records in order to obtain the necessary frequency resolution required in many applications. Furthermore, these methods suffer from spectral leakage effects, due to window-

ing, that are inherent in finite-length data records. Often, the spectral leakage masks weak signals that are present in the data.

From one point of view, the basic limitation of the nonparametric methods is the inherent assumption that the autocorrelation estimate $r_{xx}(m)$ is zero for $m \geq N$, as implied by (12.1.33). This assumption severely limits the frequency resolution and the quality of the power spectrum estimate that is achieved. From another viewpoint, the inherent assumption in the periodogram estimate is that the data are periodic with period N . Neither one of these assumptions is realistic.

In this section we describe power spectrum estimation methods that do not require such assumptions. In fact, these methods *extrapolate* the values of the autocorrelation for lags $m \geq N$. Extrapolation is possible if we have some *a priori* information on how the data were generated. In such a case a model for the signal generation can be constructed with a number of parameters that can be estimated from the observed data. From the model and the estimated parameters, we can compute the power density spectrum implied by the model.

In effect, the modeling approach eliminates the need for window functions and the assumption that the autocorrelation sequence is zero for $|m| \geq N$. As a consequence, *parametric* (model-based) power spectrum estimation methods avoid the problem of leakage and provide better frequency resolution than do the FFT-based, nonparametric methods described in the preceding section. This is especially true in applications where short data records are available due to time-variant or transient phenomena.

The parametric methods considered in this section are based on modeling the data sequence $x(n)$ as the output of a linear system characterized by a rational system function of the form

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^q b_k z^{-k}}{1 + \sum_{k=1}^p a_k z^{-k}} \quad (12.3.1)$$

The corresponding difference equation is

$$x(n) = - \sum_{k=1}^p a_k x(n-k) + \sum_{k=0}^q b_k w(n-k) \quad (12.3.2)$$

where $w(n)$ is the input sequence to the system and the observed data, $x(n)$, represents the output sequence.

In power spectrum estimation, the input sequence is not observable. However, if the observed data are characterized as a stationary random process, then the input sequence is also assumed to be a stationary random process. In such a case the power density spectrum of the data is

$$\Gamma_{xx}(f) = |H(f)|^2 \Gamma_{ww}(f)$$

where $\Gamma_{ww}(f)$ is the power density spectrum of the input sequence and $H(f)$ is the frequency response of the model.

Since our objective is to estimate the power density spectrum $\Gamma_{xx}(f)$, it is convenient to assume that the input sequence $w(n)$ is a zero-mean white noise sequence with autocorrelation

$$\gamma_{ww}(m) = \sigma_w^2 \delta(m)$$

where σ_w^2 is the variance (i.e., $\sigma_w^2 = E[|w(n)|^2]$). Then the power density spectrum of the observed data is simply

$$\Gamma_{xx}(f) = \sigma_w^2 |H(f)|^2 = \sigma_w^2 \frac{|B(f)|^2}{|A(f)|^2} \quad (12.3.3)$$

In Section 11.1 we described the representation of a stationary random process as given by (12.3.3).

In the model-based approach, the spectrum estimation procedure consists of two steps. Given the data sequence $x(n)$, $0 \leq n \leq N-1$, we estimate the parameters $\{a_k\}$ and $\{b_k\}$ of the model. Then from these estimates, we compute the power spectrum estimate according to (12.3.3).

Recall that the random process $x(n)$ generated by the pole-zero model in (12.3.1) or (12.3.2) is called an *autoregressive-moving average (ARMA) process* of order (p, q) and it is usually denoted as ARMA (p, q) . If $q = 0$ and $b_0 = 1$, the resulting system model has a system function $H(z) = 1/A(z)$ and its output $x(n)$ is called an *autoregressive (AR) process* of order p . This is denoted as AR (p) . The third possible model is obtained by setting $A(z) = 1$, so that $H(z) = B(z)$. Its output $x(n)$ is called a *moving average (MA) process* of order q and denoted as MA (q) .

Of these three linear models the AR model is by far the most widely used. The reasons are twofold. First, the AR model is suitable for representing spectra with narrow peaks (resonances). Second, the AR model results in very simple linear equations for the AR parameters. On the other hand, the MA model, as a general rule, requires many more coefficients to represent a narrow spectrum. Consequently, it is rarely used by itself as a model for spectrum estimation. By combining poles and zeros, the ARMA model provides a more efficient representation, from the viewpoint of the number of model parameters, of the spectrum of a random process.

The decomposition theorem due to Wold (1938) asserts that any ARMA or MA process can be represented uniquely by an AR model of possibly infinite order, and any ARMA or AR process can be represented by a MA model of possibly infinite order. In view of this theorem, the issue of model selection reduces to selecting the model that requires the smallest number of parameters that are also easy to compute. Usually, the choice in practice is the AR model. The ARMA model is used to a lesser extent.

Before describing methods for estimating the parameters in an AR (p) , MA (q) , and ARMA (p, q) models, it is useful to establish the basic relationships between the model parameters and the autocorrelation sequence $\gamma_{xx}(m)$. In addition, we relate the AR model parameters to the coefficients in a linear predictor for the process $x(n)$.

12.3.1 Relationships Between the Autocorrelation and the Model Parameters

In Section 11.1.2 we established the basic relationships between the autocorrelation $\{\gamma_{xx}(m)\}$ and the model parameters $\{a_k\}$ and $\{b_k\}$. For the ARMA(p, q) process, the relationship given by (11.1.18) is

$$\gamma_{xx}(m) = \begin{cases} -\sum_{k=1}^p a_k \gamma_{xx}(m-k), & m > q \\ -\sum_{k=1}^p a_k \gamma_{xx}(m-k) + \sigma_w^2 \sum_{k=0}^{q-m} h(k) b_{k+m}, & 0 \leq m \leq q \\ \gamma_{xx}^*(-m), & m < 0 \end{cases} \quad (12.3.4)$$

The relationships in (12.3.4) provide a formula for determining the model parameters $\{a_k\}$ by restricting our attention to the case $m > q$. Thus the set of linear equations

$$\begin{bmatrix} \gamma_{xx}(q) & \gamma_{xx}(q-1) & \cdots & \gamma_{xx}(q-p+1) \\ \gamma_{xx}(q+1) & \gamma_{xx}(q) & \cdots & \gamma_{xx}(q+p+2) \\ \vdots & \vdots & & \\ \gamma_{xx}(q+p-1) & \gamma_{xx}(q+p-2) & \cdots & \gamma_{xx}(q) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = - \begin{bmatrix} \gamma_{xx}(q+1) \\ \gamma_{xx}(q+2) \\ \vdots \\ \gamma_{xx}(q+p) \end{bmatrix} \quad (12.3.5)$$

can be used to solve for the model parameters $\{a_k\}$ by using estimates of the autocorrelation sequence in place of $\gamma_{xx}(m)$ for $m \geq q$. This problem is discussed in Section 12.3.8.

Another interpretation of the relationship in (12.3.5) is that the values of the autocorrelation $\gamma_{xx}(m)$ for $m > q$ are uniquely determined from the pole parameters $\{a_k\}$ and the values of $\gamma_{xx}(m)$ for $0 \leq m \leq p$. Consequently, the linear system model automatically extends the values of the autocorrelation sequence $\gamma_{xx}(m)$ for $m > p$.

If the pole parameters $\{a_k\}$ are obtained from (12.3.5), the result does not help us in determining the MA parameters $\{b_k\}$, because the equation

$$\sigma_w^2 \sum_{k=0}^{q-m} h(k) b_{k+m} = \gamma_{xx}(m) + \sum_{k=1}^p a_k \gamma_{xx}(m-k) \quad 0 \leq m \leq q$$

depends on the impulse response $h(n)$. Although the impulse response can be expressed in terms of the parameters $\{b_k\}$ by long division of $B(z)$ with the known $A(z)$, this approach results in a set of nonlinear equations for the MA parameters.

If we adopt an AR(p) model for the observed data, the relationship between the AR parameters and the autocorrelation sequence is obtained by setting $q = 0$ in (12.3.4). Thus we obtain

$$\gamma_{xx}(m) = \begin{cases} -\sum_{k=1}^p a_k \gamma_{xx}(m-k), & m > 0 \\ -\sum_{k=1}^p a_k \gamma_{xx}(m-k) + \sigma_w^2, & m = 0 \\ \gamma_{xx}^*(-m), & m < 0 \end{cases} \quad (12.3.6)$$

In this case, the AR parameters $\{a_k\}$ are obtained from the solution of the Yule-Walker or normal equations

$$\begin{bmatrix} \gamma_{xx}(0) & \gamma_{xx}(-1) & \cdots & \gamma_{xx}(-p+1) \\ \gamma_{xx}(1) & \gamma_{xx}(0) & \cdots & \gamma_{xx}(-p+2) \\ \cdots & \cdots & \ddots & \cdots \\ \gamma_{xx}(p-1) & \gamma_{xx}(p-2) & \cdots & \gamma_{xx}(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = - \begin{bmatrix} \gamma_{xx}(1) \\ \gamma_{xx}(2) \\ \vdots \\ \gamma_{xx}(p) \end{bmatrix} \quad (12.3.7)$$

and the variance σ_w^2 can be obtained from the equation

$$\sigma_w^2 = \gamma_{xx}(0) + \sum_{k=1}^p a_k \gamma_{xx}(-k) \quad (12.3.8)$$

The equations in (12.3.7) and (12.3.8) are usually combined into a single matrix equation of the form

$$\begin{bmatrix} \gamma_{xx}(0) & \gamma_{xx}(-1) & \cdots & \gamma_{xx}(-p) \\ \gamma_{xx}(1) & \gamma_{xx}(0) & \cdots & \gamma_{xx}(-p+1) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{xx}(p) & \gamma_{xx}(p-1) & \cdots & \gamma_{xx}(0) \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} \sigma_w^2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (12.3.9)$$

Since the correlation matrix in (12.3.7), or in (12.3.9), is Toeplitz, it can be efficiently inverted by use of the Levinson-Durbin algorithm.

Thus all the system parameters in the AR(p) model are easily determined from knowledge of the autocorrelation sequence $\gamma_{xx}(m)$ for $0 \leq m \leq p$. Furthermore, (12.3.6) can be used to extend the autocorrelation sequence for $m > p$, once the $\{a_k\}$ are determined.

Finally, for completeness, we indicate that in a MA(q) model for the observed data, the autocorrelation sequence $\gamma_{xx}(m)$ is related to the MA parameters $\{b_k\}$ by the equation

$$\gamma_{xx}(m) = \begin{cases} \sigma_w^2 \sum_{k=0}^q b_k b_{k+m}, & 0 \leq m \leq q \\ 0, & m > q \\ \gamma_{xx}^*(-m), & m < 0 \end{cases} \quad (12.3.10)$$

which was established in Section 11.1.

With this background established, we now describe the power spectrum estimation methods for the AR(p), ARMA(p, q), and MA(q) models.

12.3.2 The Yule–Walker Method for the AR Model Parameters

In the Yule–Walker method we simply estimate the autocorrelation from the data and use the estimates in (12.3.7) to solve for the AR model parameters. In this method it is desirable to use the biased form of the autocorrelation estimate,

$$r_{xx}(m) = \frac{1}{N} \sum_{n=0}^{N-m-1} x^*(n)x(n+m) \quad m \geq 0 \quad (12.3.11)$$

to ensure that the autocorrelation matrix is positive semidefinite. The result is a stable AR model. Although stability is not a critical issue in power spectrum estimation, it is conjectured that a stable AR model best represents the data.

The Levinson–Durbin algorithm described in Chapter 11 with $r_{xx}(m)$ substituted for $\gamma_{xx}(m)$ yields the AR parameters. The corresponding power spectrum estimate is

$$P_{xx}^{YW}(f) = \frac{\hat{\sigma}_{wp}^2}{|1 + \sum_{k=1}^p \hat{a}_p(k)e^{-j2\pi fk}|^2} \quad (12.3.12)$$

where $\hat{a}_p(k)$ are estimates of the AR parameters obtained from the Levinson–Durbin recursions and

$$\hat{\sigma}_{wp}^2 = \hat{E}_p^f = r_{xx}(0) \prod_{k=1}^p [1 - |\hat{a}_k(k)|^2] \quad (12.3.13)$$

is the estimated minimum mean-square value for the p th-order predictor. An example illustrating the frequency resolution capabilities of this estimator is given in Section 12.3.9.

In estimating the power spectrum of sinusoidal signals via AR models, Lacoss (1971) showed that spectral peaks in an AR spectrum estimate are proportional to the square of the power of the sinusoidal signal. On the other hand, the area under the peak in the power density spectrum is linearly proportional to the power of the sinusoid. This characteristic behavior holds for all AR model-based estimation methods.

12.3.3 The Burg Method for the AR Model Parameters

The method devised by Burg (1968) for estimating the AR parameters can be viewed as an order-recursive least-squares lattice method, based on the minimization of the forward and backward errors in linear predictors, with the constraint that the AR parameters satisfy the Levinson–Durbin recursion.

To derive the estimator, suppose that we are given the data $x(n)$, $n = 0, 1, \dots, N-1$, and let us consider the forward and backward linear prediction estimates of order m , given as

$$\begin{aligned}\hat{x}(n) &= -\sum_{k=1}^m a_m(k)x(n-k) \\ \hat{x}(n-m) &= -\sum_{k=1}^m a_m^*(k)x(n+k-m)\end{aligned}\quad (12.3.14)$$

and the corresponding forward and backward errors $f_m(n)$ and $g_m(n)$ defined as $f_m(n) = x(n) - \hat{x}(n)$ and $g_m(n) = x(n-m) - \hat{x}(n-m)$ where $a_m(k)$, $0 \leq k \leq m-1$, $m = 1, 2, \dots, p$, are the prediction coefficients. The least-squares error is

$$\mathcal{E}_m = \sum_{n=m}^{N-1} [|f_m(n)|^2 + |g_m(n)|^2] \quad (12.3.15)$$

This error is to be minimized by selecting the prediction coefficients, subject to the constraint that they satisfy the Levinson–Durbin recursion given by

$$\begin{aligned}a_m(k) &= a_{m-1}(k) + K_m a_{m-1}^*(m-k) & 1 \leq k \leq m-1 \\ & & 1 \leq m \leq p\end{aligned}\quad (12.3.16)$$

where $K_m = a_m(m)$ is the m th reflection coefficient in the lattice filter realization of the predictor. When (12.3.16) is substituted into the expressions for $f_m(n)$ and $g_m(n)$, the result is the pair of order-recursive equations for the forward and backward prediction errors given by (11.2.4).

Now, if we substitute from (11.2.4) into (12.3.16) and perform the minimization of $\mathcal{E}_m$ with respect to the complex-valued reflection coefficient K_m , we obtain the result

$$\hat{K}_m = \frac{-\sum_{n=m}^{N-1} f_{m-1}(n)g_{m-1}^*(n-1)}{\frac{1}{2}\sum_{n=m}^{N-1} [|f_{m-1}(n)|^2 + |g_{m-1}(n-1)|^2]} \quad m = 1, 2, \dots, p \quad (12.3.17)$$

The term in the numerator of (12.3.17) is an estimate of the crosscorrelation between the forward and backward prediction errors. With the normalization factors in the denominator of (12.3.17), it is apparent that $|K_m| < 1$, so that the all-pole model obtained from the data is stable. The reader should note the similarity of (12.3.17) to its statistical counterparts given by (11.2.29).

We note that the denominator in (12.3.17) is simply the least-squares estimate of the forward and backward errors, E_{m-1}^f and E_{m-1}^b , respectively. Hence (12.3.17)

can be expressed as

$$\hat{K}_m = \frac{-\sum_{n=m}^{N-1} f_{m-1}(n) g_{m-1}^*(n-1)}{\frac{1}{2}[\hat{E}_{m-1}^f + \hat{E}_{m-1}^b]} \quad m = 1, 2, \dots, p \quad (12.3.18)$$

where $\hat{E}_{m-1}^f + \hat{E}_{m-1}^b$ is an estimate of the total squared error E_m . We leave it as an exercise for the reader to verify that the denominator term in (12.3.18) can be computed in an order-recursive fashion according to the relation

$$\hat{E}_m = (1 - |\hat{K}_m|^2) \hat{E}_{m-1} - |f_{m-1}(m-1)|^2 - |g_{m-1}(m-2)|^2 \quad (12.3.19)$$

where $\hat{E}_m \equiv \hat{E}_m^f + \hat{E}_m^b$ is the total least-squares error. This result is due to Andersen (1978).

To summarize, the Burg algorithm computes the reflection coefficients in the equivalent lattice structure as specified by (12.3.18) and (12.3.19), and the Levinson-Durbin algorithm is used to obtain the AR model parameters. From the estimates of the AR parameters, we form the power spectrum estimate

$$P_{xx}^{BU}(f) = \frac{\hat{E}_p}{\left| 1 + \sum_{k=1}^p \hat{a}_p(k) e^{-j2\pi f k} \right|^2} \quad (12.3.20)$$

The major advantages of the Burg method for estimating the parameters of the AR model are (1) it results in high frequency resolution, (2) it yields a stable AR model, and (3) it is computationally efficient.

The Burg method is known to have several disadvantages, however. First, it exhibits spectral line splitting at high signal-to-noise ratios. [see the paper by Fougere et al. (1976)]. By line splitting, we mean that the spectrum of $x(n)$ may have a single sharp peak, but the Burg method may result in two or more closely spaced peaks. For high-order models, the method also introduces spurious peaks. Furthermore, for sinusoidal signals in noise, the Burg method exhibits a sensitivity to the initial phase of a sinusoid, especially in short data records. This sensitivity is manifest as a frequency shift from the true frequency, resulting in a phase dependent frequency bias. For more details on some of these limitations the reader is referred to the papers of Chen and Stegen (1974), Ulrych and Clayton (1976), Fougere et al. (1976), Kay and Marple (1979), Swingler (1979a, 1980), Herring (1980), and Thorvaldsen (1981).

Several modifications have been proposed to overcome some of the more important limitations of the Burg method: namely, the line splitting, spurious peaks, and frequency bias. Basically, the modifications involve the introduction of a weighting (window) sequence on the squared forward and backward errors. That is, the least-squares optimization is performed on the weighted squared

errors

$$\mathcal{E}_m^{WB} = \sum_{n=m}^{N-1} w_m(n) [|f_m(n)|^2 + |g_m(n)|^2] \quad (12.3.21)$$

which, when minimized, results in the reflection coefficient estimates

$$\hat{K}_m = \frac{\sum_{n=m}^{N-1} w_{m-1}(n) f_{m-1}(n) g_{m-1}^*(n-1)}{\frac{1}{2} \sum_{n=m}^{N-1} w_{m-1}(n) [|f_{m-1}(n)|^2 + |g_{m-1}(n-1)|^2]} \quad (12.3.22)$$

In particular, we mention the use of a Hamming window used by Swingler (1979b), a quadratic or parabolic window used by Kaveh and Lippert (1983), the energy weighting method used by Nikias and Scott (1982), and the data-adaptive energy weighting used by Helme and Nikias (1985).

These windowing and energy weighting methods have proved effective in reducing the occurrence of line splitting and spurious peaks, and are also effective in reducing frequency bias.

The Burg method for power spectrum estimation is usually associated with *maximum entropy spectrum estimation*, a criterion used by Burg (1967, 1975) as a basis for AR modeling in parametric spectrum estimation. The problem considered by Burg was how best to extrapolate from the given values of the autocorrelation sequence $\gamma_{xx}(m)$, $0 \leq m \leq p$, the values for $m > p$, such that the entire autocorrelation sequence is positive semidefinite. Since an infinite number of extrapolations are possible, Burg postulated that the extrapolations be made on the basis of maximizing uncertainty (entropy) or randomness, in the sense that the spectrum $\Gamma_{xx}(f)$ of the process is the flattest of all spectra which have the given autocorrelation values $\gamma_{xx}(m)$, $0 \leq m \leq p$. In particular the entropy per sample is proportional to the integral [see Burg (1975)]

$$\int_{-1/2}^{1/2} \ln \Gamma_{xx}(f) df \quad (12.3.23)$$

Burg found that the maximum of this integral subject to the $(p+1)$ constraints

$$\int_{-1/2}^{1/2} \Gamma_{xx}(f) e^{j2\pi fm} df = \gamma_{xx}(m) \quad 0 \leq m \leq p \quad (12.3.24)$$

is the AR(p) process for which the given autocorrelation sequence $\gamma_{xx}(m)$, $0 \leq m \leq p$ is related to the AR parameters by the equation (12.3.6). This solution provides an additional justification for the use of the AR model in power spectrum estimation.

In view of Burg's basic work in maximum entropy spectral estimation, the Burg power spectrum estimation procedure is often called the *maximum entropy method* (MEM). We should emphasize, however, that the maximum entropy spectrum is identical to the AR-model spectrum only when the exact autocorrelation

$\gamma_{xx}(m)$ is known. When only an estimate of $\gamma_{xx}(m)$ is available for $0 \leq m \leq p$, the AR-model estimates of Yule-Walker and Burg are not maximum entropy spectral estimates. The general formulation for the maximum entropy spectrum based on estimates of the autocorrelation sequence results in a set of nonlinear equations. Solutions for the maximum entropy spectrum with measurement errors in the correlation sequence have been obtained by Newman (1981) and Schott and McClellan (1984).

12.3.4 Unconstrained Least-Squares Method for the AR Model Parameters

As described in the preceding section, the Burg method for determining the parameters of the AR model is basically a least-squares lattice algorithm with the added constraint that the predictor coefficients satisfy the Levinson recursion. As a result of this constraint, an increase in the order of the AR model requires only a single parameter optimization at each stage. In contrast to this approach, we may use an unconstrained least-squares algorithm to determine the AR parameters.

To elaborate, we form the forward and backward linear prediction estimates and their corresponding forward and backward errors as indicated in (12.3.14) and (12.3.15). Then we minimize the sum of squares of both errors, that is,

$$\begin{aligned} \mathcal{E}_p &= \sum_{n=p}^{N-1} [|f_p(n)|^2 + |g_p(n)|^2] \\ &= \sum_{n=p}^{N-1} \left[\left| x(n) + \sum_{k=1}^p a_p(k)x(n-k) \right|^2 + \left| x(n-p) + \sum_{k=1}^p a_p^*(k)x(n+k-p) \right|^2 \right] \end{aligned} \quad (12.3.25)$$

which is the same performance index as in the Burg method. However, we do not impose the Levinson-Durbin recursion in (12.3.25) for the AR parameters. The unconstrained minimization of $\mathcal{E}_p$ with respect to the prediction coefficients yields the set of linear equations

$$\sum_{k=1}^p a_p(k)r_{xx}(l, k) = -r_{xx}(l, 0) \quad l = 1, 2, \dots, p \quad (12.3.26)$$

where, by definition, the autocorrelation $r_{xx}(l, k)$ is

$$r_{xx}(l, k) = \sum_{n=p}^{N-1} [x(n-k)x^*(n-l) + x(n-p+l)x^*(n-p+k)] \quad (12.3.27)$$

The resulting residual least-squares error is

$$\mathcal{E}_p^{LS} = r_{xx}(0, 0) + \sum_{k=1}^p \hat{a}_p(k)r_{xx}(0, k) \quad (12.3.28)$$

Hence the unconstrained least-squares power spectrum estimate is

$$P_{xx}^{LS}(f) = \frac{\xi_p^{LS}}{\left| 1 + \sum_{k=1}^p \hat{a}_p(k) e^{-j2\pi f k} \right|^2} \quad (12.3.29)$$

The correlation matrix in (12.3.27), with elements $r_{xx}(l, k)$, is not Toeplitz, so that the Levinson–Durbin algorithm cannot be applied. However, the correlation matrix has sufficient structure to make it possible to devise computationally efficient algorithms with computational complexity proportional to p^2 . Marple (1980) devised such an algorithm, which has a lattice structure and employs Levinson–Durbin-type order recursions and additional time recursions.

This form of the unconstrained least-squares method described has also been called the *unwindowed data* least-squares method. It has been proposed for spectrum estimation in several papers, including the papers by Burg (1967), Nuttall (1976), and Ulrych and Clayton (1976). Its performance characteristics have been found to be superior to the Burg method, in the sense that the unconstrained least-squares method does not exhibit the same sensitivity to such problems as line splitting, frequency bias, and spurious peaks. In view of the computational efficiency of Marple’s algorithm, which is comparable to the efficiency of the Levinson–Durbin algorithm, the unconstrained least-squares method is very attractive. With this method there is no guarantee that the estimated AR parameters yield a stable AR model. However, in spectrum estimation, this is not considered to be a problem.

12.3.5 Sequential Estimation Methods for the AR Model Parameters

The three power spectrum estimation methods described in the preceding sections for the AR model can be classified as block processing methods. These methods obtain estimates of the AR parameters from a block of data, say $x(n)$, $n = 0, 1, \dots, N-1$. The AR parameters, based on the block of N data points, are then used to obtain the power spectrum estimate.

In situations where data are available on a continuous basis, we can still segment the data into blocks of N points and perform spectrum estimation on a block-by-block basis. This is often done in practice, for both real-time and non-real-time applications. However, in such applications, there is an alternative approach based on sequential (in time) estimation of the AR model parameters as each new data point becomes available. By introducing a weighting function into past data samples, it is possible to deemphasize the effect of older data samples as new data are received.

Sequential lattice methods based on recursive least squares can be employed to optimally estimate the prediction and reflection coefficients in the lattice realization of the forward and backward linear predictors. The recursive equa-

tions for the prediction coefficients relate directly to the AR model parameters. In addition to the order-recursive nature of these equations, as implied by the lattice structure, we can also obtain time-recursive equations for the reflection coefficients in the lattice and for the forward and backward prediction coefficients.

Sequential recursive least-squares algorithms are equivalent to the unconstrained least-squares, block processing method described in the preceding section. Hence the power spectrum estimates obtained by the sequential recursive least-squares method retain the desirable properties of the block processing algorithm described in Section 12.3.4. Since the AR parameters are being continuously estimated in a sequential estimation algorithm, power spectrum estimates can be obtained as often as desired, from once per sample to once every N samples. By properly weighting past data samples, the sequential estimation methods are particularly suitable for estimating and tracking time-variant power spectra resulting from nonstationary signal statistics.

The computational complexity of sequential estimation methods is generally proportional to p , the order of the AR process. As a consequence, sequential estimation algorithms are computationally efficient and, from this viewpoint, may offer some advantage over the block processing methods.

There are numerous references on sequential estimation methods. The papers by Griffiths (1975), Friedlander (1982a, b), and Kalouptsidis and Theodoridis (1987) are particularly relevant to the spectrum estimation problem.

12.3.6 Selection of AR Model Order

One of the most important aspects of the use of the AR model is the selection of the order p . As a general rule, if we select a model with too low an order, we obtain a highly smoothed spectrum. On the other hand, if p is selected too high, we run the risk of introducing spurious low-level peaks in the spectrum. We mentioned previously that one indication of the performance of the AR model is the mean-square value of the residual error, which, in general, is different for each of the estimators described above. The characteristic of this residual error is that it decreases as the order of the AR model is increased. We can monitor the rate of decrease and decide to terminate the process when the rate of decrease becomes relatively slow. It is apparent, however, that this approach may be imprecise and ill-defined, and other methods should be investigated.

Much work has been done by various researchers on this problem and many experimental results have been given in the literature [e.g., the papers by Gersch and Sharpe (1973), Ulrych and Bishop (1975), Tong (1975, 1977), Jones (1976), Nuttall (1976), Berryman (1978), Kaveh and Bruzzone (1979), and Kashyap (1980)].

Two of the better known criteria for selecting the model order have been proposed by Akaike (1969, 1974). With the first, called the *final prediction error*

(FPE) criterion, the order is selected to minimize the performance index

$$\text{FPE}(p) = \hat{\sigma}_{wp}^2 \left(\frac{N + p + 1}{N - p - 1} \right) \quad (12.3.30)$$

where $\hat{\sigma}_{wp}^2$ is the estimated variance of the linear prediction error. This performance index is based on minimizing the mean-square error for a one-step predictor.

The second criterion proposed by Akaike (1974), called the *Akaike information criterion* (AIC), is based on selecting the order that minimizes

$$\text{AIC}(p) = \ln \hat{\sigma}_{wp}^2 + 2p/N \quad (12.3.31)$$

Note that the term $\hat{\sigma}_{wp}^2$ decreases and therefore $\ln \hat{\sigma}_{wp}^2$ also decreases as the order of the AR model is increased. However, $2p/N$ increases with an increase in p . Hence a minimum value is obtained for some p .

An alternative information criterion, proposed by Rissanen (1983), is based on selecting the order that *minimizes the description length* (MDL), where MDL is defined as

$$\text{MDL}(p) = N \ln \hat{\sigma}_{wp}^2 + p \ln N \quad (12.3.32)$$

A fourth criterion has been proposed by Parzen (1974). This is called the *criterion autoregressive transfer* (CAT) function and is defined as

$$\text{CAT}(p) = \left(\frac{1}{N} \sum_{k=1}^p \frac{1}{\hat{\sigma}_{wk}^2} \right) - \frac{1}{\hat{\sigma}_{wp}^2} \quad (12.3.33)$$

where

$$\hat{\sigma}_{wk}^2 = \frac{N}{N-k} \hat{\sigma}_{wk}^2 \quad (12.3.34)$$

The order p is selected to minimize $\text{CAT}(p)$.

In applying this criteria, the mean should be removed from the data. Since $\hat{\sigma}_{wk}^2$ depends on the type of spectrum estimate we obtain, the model order is also a function of the criterion.

The experimental results given in the references just cited indicate that the model-order selection criteria do not yield definitive results. For example, Ulrych and Bishop (1975), Jones (1976), and Berryman (1978), found that the $\text{FPE}(p)$ criterion tends to underestimate the model order. Kashyap (1980) showed that the AIC criterion is statistically inconsistent as $N \rightarrow \infty$. On the other hand, the MDL information criterion proposed by Rissanen is statistically consistent. Other experimental results indicate that for small data lengths, the order of the AR model should be selected to be in the range $N/3$ to $N/2$ for good results. It is apparent that in the absence of any prior information regarding the physical process that resulted in the data, one should try different model orders and different criteria and, finally, consider the different results.

12.3.7 MA Model for Power Spectrum Estimation

As shown in Section 12.3.1, the parameters in a MA(q) model are related to the statistical autocorrelation $\gamma_{xx}(m)$ by (12.3.10). However,

$$B(z)B(z^{-1}) = D(z) = \sum_{m=-q}^q d_m z^{-m} \quad (12.3.35)$$

where the coefficients $\{d_m\}$ are related to the MA parameters by the expression

$$d_m = \sum_{k=0}^{q-|m|} b_k b_{k+m} \quad |m| \leq q \quad (12.3.36)$$

Clearly, then,

$$\gamma_{xx}(m) = \begin{cases} \sigma_w^2 d_m, & |m| \leq q \\ 0, & |m| > q \end{cases} \quad (12.3.37)$$

and the power spectrum for the MA(q) process is

$$\Gamma_{xx}^{\text{MA}}(f) = \sum_{m=-q}^q \gamma_{xx}(m) e^{-j2\pi f m} \quad (12.3.38)$$

It is apparent from these expressions that we do not have to solve for the MA parameters $\{b_k\}$ to estimate the power spectrum. The estimates of the autocorrelation $\gamma_{xx}(m)$ for $|m| \leq q$ suffice. From such estimates we compute the estimated MA power spectrum, given as

$$P_{xx}^{\text{MA}}(f) = \sum_{m=-q}^q r_{xx}(m) e^{-j2\pi f m} \quad (12.3.39)$$

which is identical to the classical (nonparametric) power spectrum estimate described in Section 12.1.

There is an alternative method for determining $\{b_k\}$ based on a high-order AR approximation to the MA process. To be specific, let the MA(q) process be modeled by an AR(p) model, where $p \gg q$. Then $B(z) = 1/A(z)$, or equivalently, $B(z)A(z) = 1$. Thus the parameters $\{b_k\}$ and $\{a_k\}$ are related by a convolution sum, which can be expressed as

$$\hat{a}_n + \sum_{k=1}^q b_k \hat{a}_{n-k} = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (12.3.40)$$

where $\{\hat{a}_n\}$ are the parameters obtained by fitting the data to an AR(p) model.

Although this set of equations can be easily solved for the $\{b_k\}$, a better fit is obtained by using a least-squares error criterion. That is, we form the squared error

$$\mathcal{E} = \sum_{n=0}^p \left[\hat{a}_n + \sum_{k=1}^q b_k \hat{a}_{n-k} \right]^2 - 1 \quad \hat{a}_0 = 1, \quad \hat{a}_k = 0, \quad k < 0 \quad (12.3.41)$$

which is minimized by selecting the MA(q) parameters $\{b_k\}$. The result of this

minimization is

$$\hat{\mathbf{b}} = -\mathbf{R}_{aa}^{-1} \mathbf{r}_{aa} \quad (12.3.42)$$

where the elements of $\mathbf{R}_{aa}$ and $\mathbf{r}_{aa}$ are given as

$$R_{aa}(|i-j|) = \sum_{n=0}^{p-|i-j|} \hat{a}_n \hat{a}_{n+|i-j|} \quad i, j = 1, 2, \dots, q \quad (12.3.43)$$

$$r_{aa}(i) = \sum_{n=0}^{p-i} \hat{a}_n \hat{a}_{n+i} \quad i = 1, 2, \dots, q$$

This least squares method for determining the parameters of the MA(q) model is attributed to Durbin (1959). It has been shown by Kay (1988) that this estimation method is approximately the maximum likelihood under the assumption that the observed process is Gaussian.

The order q of the MA model may be determined empirically by several methods. For example, the AIC for MA models has the same form as for AR models,

$$\text{AIC}(q) = \ln \sigma_{wq}^2 + \frac{2q}{N} \quad (12.3.44)$$

where σ_{wq}^2 is an estimate of the variance of the white noise. Another approach, proposed by Chow (1972b), is to filter the data with the inverse MA(q) filter and test the filtered output for whiteness.

12.3.8 ARMA Model for Power Spectrum Estimation

The Burg algorithm, its variations, and the least-squares method described in the previous sections provide reliable high-resolution spectrum estimates based on the AR model. An ARMA model provides us with an opportunity to improve on the AR spectrum estimate, perhaps, by using fewer model parameters.

The ARMA model is particularly appropriate when the signal has been corrupted by noise. For example, suppose that the data $x(n)$ are generated by an AR system, where the system output is corrupted by additive white noise. The z -transform of the autocorrelation of the resultant signal can be expressed as

$$\begin{aligned} \Gamma_{xx}(z) &= \frac{\sigma_w^2}{A(z)A(z^{-1})} + \sigma_n^2 \\ &= \frac{\sigma_w^2 + \sigma_n^2 A(z)A(z^{-1})}{A(z)A(z^{-1})} \end{aligned} \quad (12.3.45)$$

where σ_n^2 is the variance of the additive noise. Therefore, the process $x(n)$ is ARMA(p, p), where p is the order of the autocorrelation process. This relationship provides some motivation for investigating ARMA models for power spectrum estimation.

As we have demonstrated in Section 12.3.1, the parameters of the ARMA model are related to the autocorrelation by the equation in (12.3.4). For lags

$|m| > q$, the equation involves only the AR parameters $\{a_k\}$. With estimates substituted in place of $\gamma_{xx}(m)$, we can solve the p equations in (12.3.5) to obtain $\hat{a}_k$. For high-order models, however, this approach is likely to yield poor estimates of the AR parameters due to the poor estimates of the autocorrelation for large lags. Consequently, this approach is not recommended.

A more reliable method is to construct an overdetermined set of linear equations for $m > q$, and to use the method of least squares on the set of overdetermined equations, as proposed by Cadzow (1979). To elaborate, suppose that the autocorrelation sequence can be accurately estimated up to lag M , where $M > p + q$. Then we can write the following set of linear equations:

$$\begin{bmatrix} r_{xx}(q) & r_{xx}(q-1) & \cdots & r_{xx}(q-p+1) \\ r_{xx}(q+1) & r_{xx}(q) & \cdots & r_{xx}(q-p+2) \\ \vdots & \vdots & & \vdots \\ r_{xx}(M-1) & r_{xx}(M-2) & \cdots & r_{xx}(M-p) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = - \begin{bmatrix} r_{xx}(q+1) \\ r_{xx}(q+2) \\ \vdots \\ r_{xx}(M) \end{bmatrix} \quad (12.3.46)$$

or equivalently,

$$\mathbf{R}_{xx} \mathbf{a} = -\mathbf{r}_{xx} \quad (12.3.47)$$

Since $\mathbf{R}_{xx}$ is of dimension $(M-q) \times p$, and $M-q > p$ we can use the least-squares criterion to solve for the parameter vector $\mathbf{a}$. The result of this minimization is

$$\hat{\mathbf{a}} = -(\mathbf{R}_{xx}' \mathbf{R}_{xx})^{-1} \mathbf{R}_{xx}' \mathbf{r}_{xx} \quad (12.3.48)$$

This procedure is called the *least-squares modified Yule-Walker method*. A weighting factor can also be applied to the autocorrelation sequence to deemphasize the less reliable estimates for large lags.

Once the parameters for the AR part of the model have been estimated as indicated above, we have the system

$$\hat{A}(z) = 1 + \sum_{k=1}^p \hat{a}_k z^{-k} \quad (12.3.49)$$

The sequence $x(n)$ can now be filtered by the FIR filter $\hat{A}(z)$ to yield the sequence

$$v(n) = x(n) + \sum_{k=1}^p \hat{a}_k x(n-k) \quad n = 0, 1, \dots, N-1 \quad (12.3.50)$$

The cascade of the ARMA(p, q) model with $\hat{A}(z)$ is approximately the MA(q) process generated by the model $B(z)$. Hence we can apply the MA estimate given in the preceding section to obtain the MA spectrum. To be specific, the filtered sequence $v(n)$ for $p \leq n \leq N-1$ is used to form the estimated correlation sequences $r_{vv}(m)$, from which we obtain the MA spectrum

$$P_{vv}^{\text{MA}}(f) = \sum_{m=-q}^q r_{vv}(m) e^{-j2\pi f m} \quad (12.3.51)$$

First, we observe that the parameters $\{b_k\}$ are not required to determine the power spectrum. Second, we observe that $r_{vv}(m)$ is an estimate of the autocorrelation for the MA model given by (12.3.10). In forming the estimate $r_{vv}(m)$, weighting (e.g., with the Bartlett window) may be used to deemphasize correlation estimates for large lags. In addition, the data may be filtered by a backward filter, thus creating another sequence, say $v^b(n)$, so that both $v(n)$ and $v^b(n)$ can be used in forming the estimate of the autocorrelation $r_{vv}(m)$, as proposed by Kay (1980). Finally, the estimated ARMA power spectrum is

$$\hat{P}_{xx}^{\text{ARMA}}(f) = \frac{P_{vv}^{\text{MA}}(f)}{\left| 1 + \sum_{k=1}^p \hat{a}_k e^{-j2\pi f k} \right|^2} \quad (12.3.52)$$

The problem of order selection for the ARMA(p, q) model has been investigated by Chow (1972a, b) and Bruzzone and Kaveh (1980). For this purpose the minimum of the AIC index

$$\text{AIC}(p, q) = \ln \hat{\sigma}_{wpg}^2 + \frac{2(p+q)}{N} \quad (12.3.53)$$

can be used, where $\hat{\sigma}_{wpg}^2$ is an estimate of the variance of the input error. An additional test on the adequacy of a particular ARMA(p, q) model is to filter the data through the model and test for whiteness of the output data. This requires that the parameters of the MA model be computed from the estimated autocorrelation, using spectral factorization to determine $B(z)$ from $D(z) = B(z)B(z^{-1})$.

For additional reading on ARMA power spectrum estimation, the reader is referred to the papers by Graupe et al. (1975), Cadzow (1981, 1982), Kay (1980), and Friedlander (1982b).

12.3.9 Some Experimental Results

In this section we present some experimental results on the performance of AR and ARMA power spectrum estimates obtained by using artificially generated data. Our objective is to compare the spectral estimation methods on the basis of their frequency resolution, bias, and their robustness in the presence of additive noise.

The data consist of either one or two sinusoids and additive Gaussian noise. The two sinusoids are spaced Δf apart. Clearly, the underlying process is ARMA(4, 4). The results that are shown employ an AR(p) model for these data. For high signal-to-noise ratios (SNRs) we expect the AR(4) to be adequate. However, for low SNRs, a higher-order AR model is needed to approximate the ARMA(4, 4) process. The results given below are consistent with this statement. The SNR is defined as $10 \log_{10} A^2/2\sigma^2$, where σ^2 is variance of the additive noise and A is the amplitude of the sinusoid.

In Fig. 12.6 we illustrate the results for $N = 20$ data points based on an AR(4) model with a SNR = 20 dB and $\Delta f = 0.13$. Note that the Yule-Walker method gives an extremely smooth (broad) spectral estimate with small peaks. If Δf is

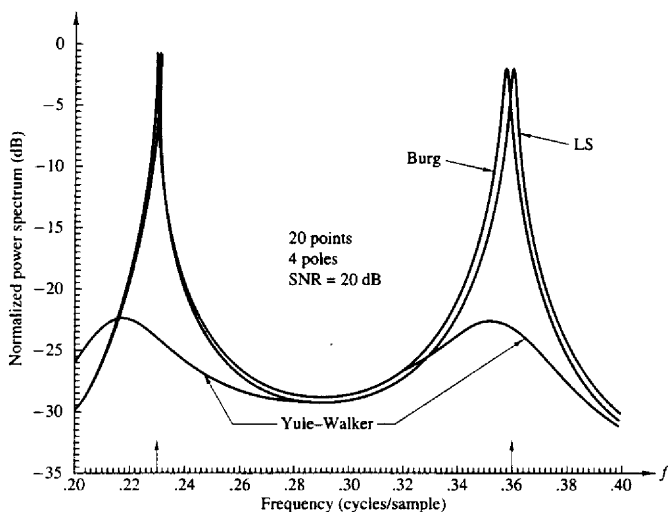


Figure 12.6 Comparison of AR spectrum estimation methods.

decreased to $\Delta f = 0.07$, the Yule-Walker method no longer resolves the peaks as illustrated in Fig. 12.7. Some bias is also evident in the Burg method. Of course, by increasing the number of data points the Yule-Walker method eventually is able to resolve the peaks. However, the Burg and least-squares methods are clearly superior for short data records.

The effect of additive noise on the estimate is illustrated in Fig. 12.8 for the least-squares method. The effect of filter order on the Burg and least-squares methods is illustrated in Figs. 12.9 and 12.10, respectively. Both methods exhibit spurious peaks as the order is increased to $p = 12$.

The effect of initial phase is illustrated in Figs. 12.11 and 12.12 for the Burg and least-squares methods. It is clear that the least-squares method exhibits less sensitivity to initial phase than the Burg algorithm.

An example of line splitting for the Burg method is shown in Fig. 12.13 with $p = 12$. It does not occur for the AR(8) model. The least-squares method did not exhibit line splitting under the same conditions. On the other hand, the line splitting on the Burg method disappeared with an increase in the number of data points N .

Figures 12.14 and 12.15 illustrate the resolution properties of the Burg and least-squares methods for $\Delta f = 0.07$ and $N = 20$ points at low SNR (3 dB). Since the additive noise process is ARMA, a higher-order AR model is required to provide a good approximation at low SNR. Hence the frequency resolution improves as the order is increased.

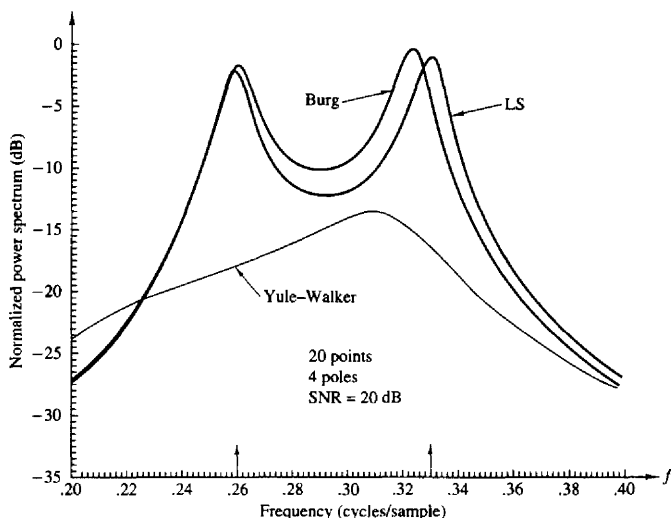


Figure 12.7 Comparison of AR spectrum estimation methods.

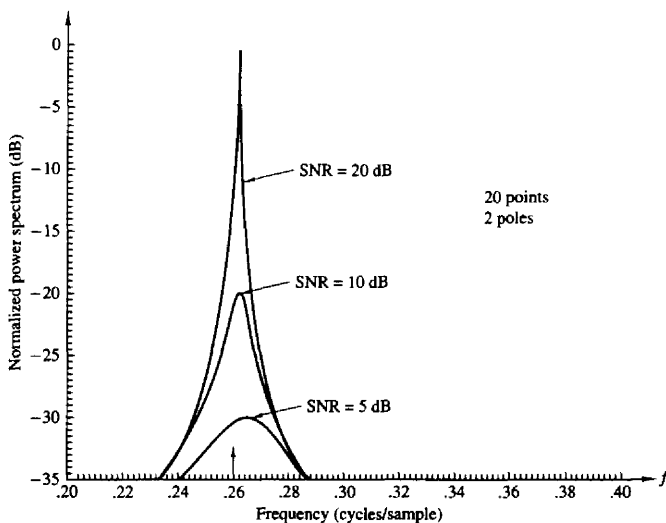


Figure 12.8 Effect of additive noise on LS method.

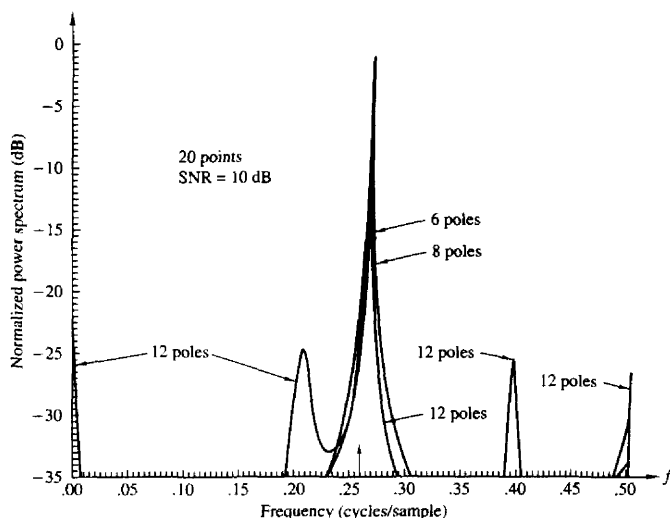


Figure 12.9 Effect of filter order of Burg method.

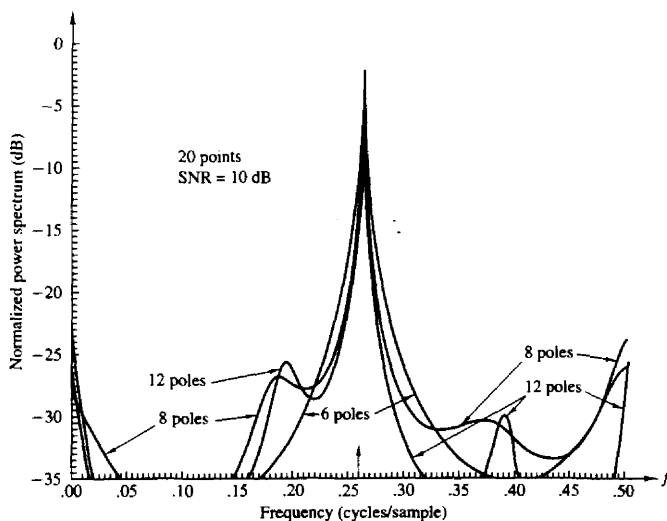


Figure 12.10 Effect of filter order on LS method.

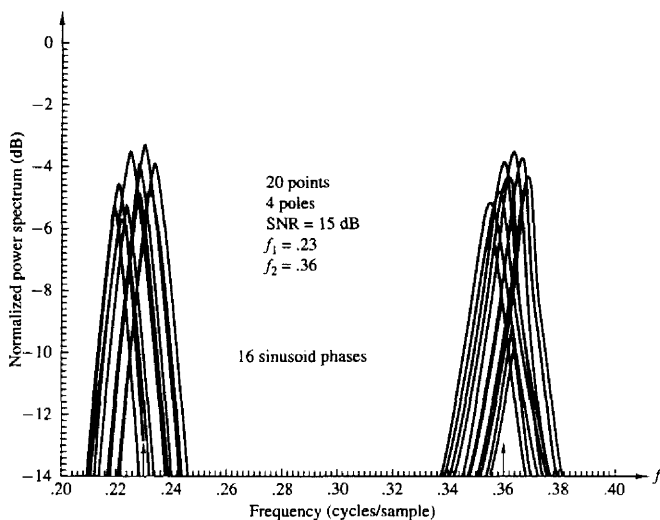


Figure 12.11 Effect of initial phase on Burg method.

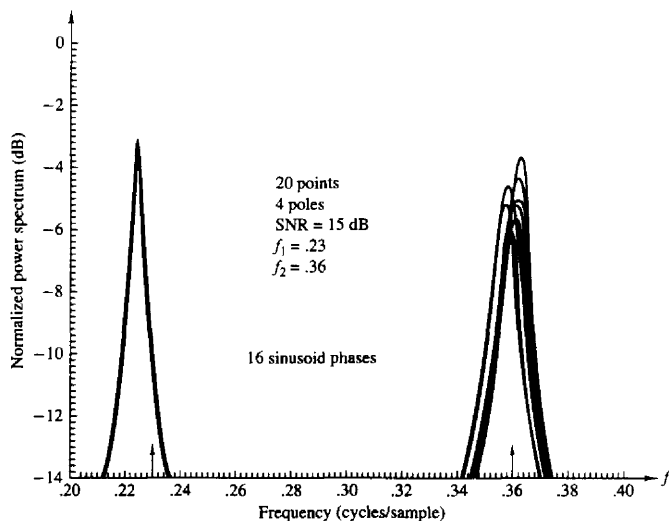


Figure 12.12 Effect of initial phase on LS method.

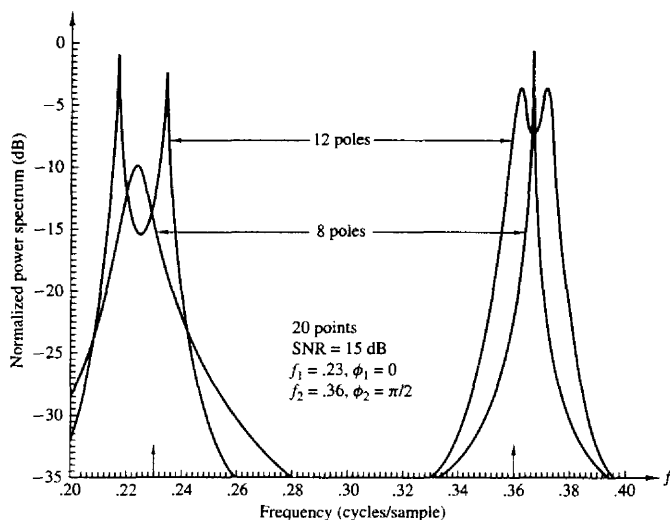
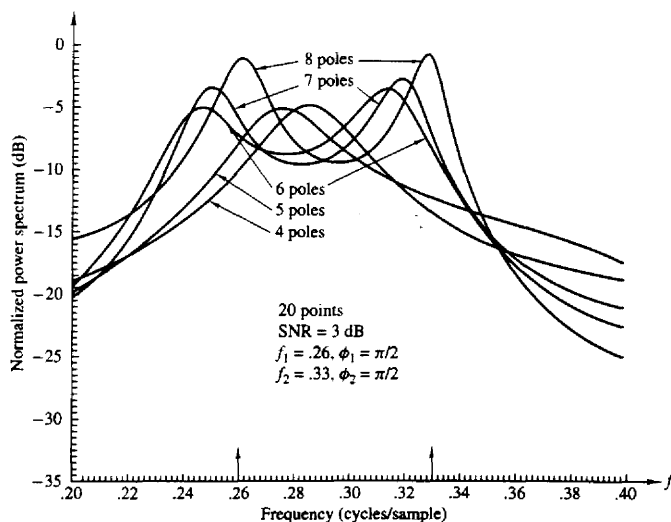


Figure 12.13 Line splitting in Burg method.


 Figure 12.14 Frequency resolution of Burg method with $N = 20$ points.

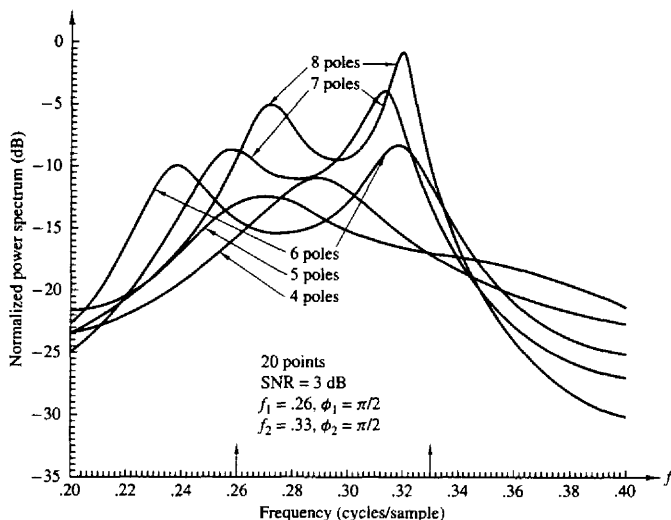


Figure 12.15 Frequency resolution of LS method with $N = 20$ points.

The FPE for the Burg method is illustrated in Fig. 12.16 for an SNR = 3 dB. For this SNR the optimum value is $p = 12$ according to the FPE criterion.

The Burg and least-squares methods were also tested with data from a narrowband process, obtained by exciting a four-pole (two pairs of complex-conjugate poles) narrowband filter and selecting a portion of the output sequence for the data record. Figure 12.17 illustrates the superposition of 20 data records of 20 points each. We observe a relatively small variability. In contrast, the Burg method exhibited a much larger variability, approximately a factor of 2 compared to the least-squares method. The results shown in Figs. 12.6 through 12.17 are taken from Poole (1981).

Finally, we show in Fig. 12.18 the ARMA(10, 10) spectral estimates obtained by Kay (1980) for two sinusoids in noise using the least-squares ARMA method described in Section 12.3.8, as an illustration of the quality of power spectrum estimation obtained with the ARMA model.

12.4 MINIMUM VARIANCE SPECTRAL ESTIMATION

The spectral estimator proposed by Capon (1969) was intended for use in large seismic arrays for frequency-wave number estimation. It was later adapted to single-time-series spectrum estimation by Lacoss (1971), who demonstrated that

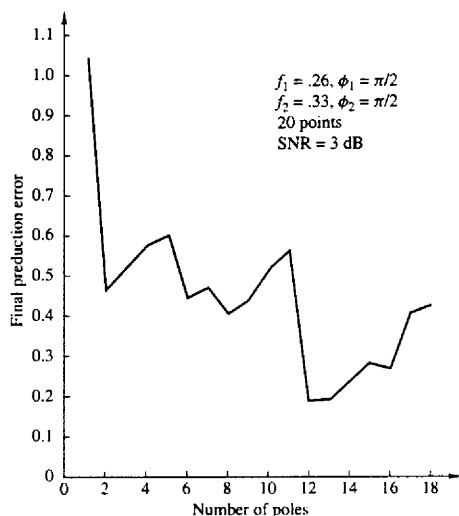


Figure 12.16 Final prediction error for Burg estimate.

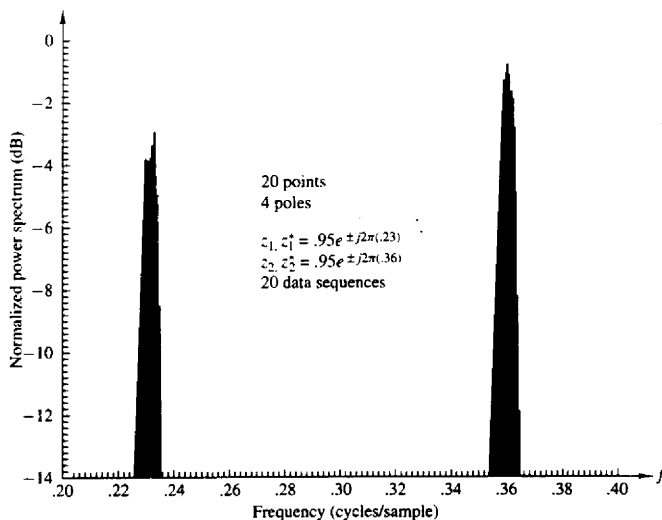


Figure 12.17 Effect of starting point in sequence on LS method.